Financial Regulation, Pension Investment, and Economic Growth

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Abstract

This paper asks how changes in investors' risk-taking capacity affect asset prices and economic growth. I develop a Schumpeterian growth model with risk-averse investors. Risk limits raise the market risk premium, reduce R&D, and can dampen entry innovation in general equilibrium when the rise in the risk premium is sufficiently strong. Using a tightening of risk requirements on British pension funds as a natural experiment, I show that their subsequent divestment from equities led to a persistent fall in stock prices among firms more exposed to pension investors pre-reform. In response, these firms cut R&D expenditure and reduced long-term investment. Counterfactual simulations using a calibrated version of my model suggest that the impact of risk limits on growth can be sizeable and persistent when equity markets are inelastic.

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1 Introduction

Since the Global Financial Crisis, economic growth in many advanced economies has stalled. A frequently-cited explanation for this slowdown is that tighter financial regulation has increased levels of de-facto risk-aversion among many investors (e.g. Draghi, 2024).

From a macroeconomic point of view, our understanding of how financial regulation can alter firms' investment incentives through its impact on investor risk aversion remains limited. When capital markets are imperfect and there are limits to arbitrage, for example due to segmentation (Gromb and Vayanos, 2010; Shleifer and Vishny, 1998) or inelastic demand (Gabaix and Koijen, 2021), tighter regulation of market participants' risk-taking capacity can decrease market valuations, increase firms' cost of capital, and thus reduce their scope to invest. If these shifts are large and persistent, financial regulation may have unanticipated consequences for growth.

In this paper, I study theoretically and empirically how regulation that reduces investors' willingness to take risk impacts economic growth. The paper is structured around three contributions. First, I introduce an endogenous growth framework with segmented equity markets and risk-averse investors. I use this model to study theoretically how limits to investor risk-taking capacity affect firms' investment incentives, and how, in general equilibrium, they shape firm dynamics and economic growth. I show that risk limits raise risk premia and reduce incumbent innovation, but need not lower growth. This is because the reduction in incumbent investment reduces the price pressure in factor input markets and allows for the reallocation of resources towards potentially more productive outside innovators. Second, I provide empirical support for my framework by leveraging a reform in the United Kingdom that tightened risk requirements on pension funds and led to a large divestment from equity markets. Consistent with the limits to arbitrage view of capital markets, I show that firms more reliant on pension investors saw a persistent fall in stock prices after the reform. In response, these firms cut back on their long-term investment. Third, I calibrate the model to quantify how a tightening of risk requirements on pension funds might have affected economic growth. Simulations suggest that the impact can be seizable when firm investment is sensitive to changes in market valuations and when equity markets are inelastic.

The starting point for the analysis is a workhorse Schumpeterian growth model with innovation by entrant and incumbent firms. To evaluate how changes in investor risk-taking capacity affect growth, I extend the model by introducing risk-averse investors who fund incumbent firms in *segmented* public equity markets. Like in the standard model, outside innovators are funded by inside equity. The model's main mechanism arises from the interaction of incumbent firms, of which there is a finite discrete number, with two types of risk-averse investors in these segmented equity markets. The first type, market-driven investors, solve a conventional portfolio problem and allocate funding based on Sharpe ratios. In contrast, risk-constrained investors, who can be thought of as defined-benefit pension funds or insurance companies, solve a mean-variance problem subject to a value-at-risk constraint that limits their exposure to volatile stocks. Given investor demand, non-financial firms invest in innovation to maximise their stock price and are subject to aggregate risk. Risk-averse investors need to be compensated to hold stocks via a risk premium. In equilibrium, firms' valuations and therefore their investment decisions depend on risk-constrained investors' capacity to take risk and thus their regulatory constraint.

The main comparative statics exercise is a tightening of risk-constrained investors' value-at-risk constraint, prompting them to sell volatile stocks. For the stock market to clear, market-driven investors need to be compensated with a higher risk premium to absorb risk-constrained investors' stock sales. Incumbent firms, faced with a higher risk premium and thus lower stock prices, cut back on innovation investment as the marginal value of growing the firms' size (the stock price) falls. In partial equilibrium, the economy's growth rate falls. In general equilibrium, the reduction in incumbent innovation has two opposing effects on entry: While a higher risk premium reduces the value of entry for outside innovators, lower incumbent investment depresses factor prices and thus the cost of entry. Hence, whether there is more or less entry in response to a rise in risk premia depends on the relative strength of these *risk premium* and *reallocation effects*. When the former dominates, the growth rate is decreasing in market risk aversion. When the latter dominates, the relationship between growth and market risk aversion, and therefore the degree of regulation, is hump-shaped. On the left side of the hump, tighter risk requirements lead to *more* growth. Theoretically, the hump-shaped relationship between regulation and growth implies the existence of an interior optimal degree of regulation on the size of investors' equity positions. This regulation level balances the relative contribution of incumbent and entrant innovators to growth. I derive an analytical solution in terms of model primitives and show that optimal regulation rises with the productivity gap between entrants and incumbents and the size of the financial sector.

The second part of the paper provides empirical evidence on the impact of risk-driven stock market divestment on firm outcomes. The setting is a natural experiment in the United Kingdom, where the collapse of a large corporate pension scheme in the late 1990s prompted the government to tighten risk requirements on defined-benefit pension schemes. Under the Pensions Act 2004, the regulator was tasked with monitoring schemes' *funding level*, defined as the difference between the market value of assets and future pension obligations. The present value of these obligations was to be calculated using yields on inflation-linked gilts. Since a deterioration of their funding level could trigger regulatory intervention, the reform incentivised schemes to sell equities and buy long-maturity bonds, thus reducing both the volatility of their assets and duration risk from their long-dated liabilities. Within three years of the reform, pension funds sold more than £65bn in equities. Using cross-sectional data on pension schemes' funding level, I show that schemes with lower funding levels pre-reform subsequently sold a larger proportion of their equity holdings.

Building on these findings, I use the reform as a natural experiment to analyse how pension funds' withdrawal from equity markets influenced the investment decisions of those non-financial firms that the funds were previously invested in. To study the impact of the reform, I introduce a new dataset of hand-collected pension fund stock holdings for a large number of defined-benefit pension schemes.

The observation that more underfunded pension schemes sold larger quantities of stocks in response to the reform motivates an event-study shift-share design. I construct an exposure measure using the fraction of pension fund stock ownership in each firm as the *share* and the subsequent cumulative divestment from equities at the fund level as the exogenous *shift*. This design exploits the idea that a pension scheme's total divestment from equities is exogenous with respect to a particular firm's characteristics, although the divestment from this firm's stock is, of course, not. The underlying assumption is that the match between pension funds and firms prior to the reform is uncorrelated with differences in post-reform firm outcomes, after controlling for observables. I provide evidence for this assumption using three strategies. First, return regressions show that the cross-sectional variation in funding levels was driven by the performance of British equities. After controlling for equity exposure, past returns are not predictive of funding levels. Second, measures of portfolio composition show that within equities, there is little compositional variation, as pension funds closely track the benchmark. Third, I corroborate my results using an alternative instrument based on schemes' membership structure, which leverages the idea that schemes with a higher ratio of pensioners to contributors should be more underfunded because of higher future pension obligations and therefore should react more sensitively to the tighter funding requirements.

My empirical analysis finds a negative effect of pension capital withdrawal on firm outcomes. I estimate that a one percentage point reduction in pension fund equity investment reduces stock prices by 0.45 percent after one year, implying that the divestment was only partially offset by other investors. This has consequences for firms' real investment choices. For every one percentage point reduction in pension investment, firms cut their capital and

R&D expenditure by 1 percent and 1.2 percent, respectively, and pivot towards more short-term investments. Here, an additional one percentage point reduction in pension investment lowers the share of long-term investment by 0.4 percent, as proxied by the asset composition of firms' balance sheets. This investment response changes firms fundamentals and leads to a further drop in stock prices. After five years, valuations are 1.75 percent lower relative to pre-reform levels.

In the final part of the paper, I quantify how the British pension reform has affected growth. I calibrate the model to a series of key pre-reform macro moments and the empirically estimated firm-level elasticity of R&D spending with respect to the funding shock, which is a sufficient statistic for the reaction of incumbent investment to the capitalisation level of the equity market. The calibration suggests that the reallocation effect on entry is relatively weak, and thus the empirically relevant case is the one where the growth rate is monotonically decreasing in the degree of regulation. To assess the impact of the reform, I then run the following exercise: holding constant all other parameters at their pre-reform level, I vary the tightness of the regulatory constraint to reproduce the observed decline in defined-benefit funds' equity portfolio share from 2002 to 2007. In the simulation, the growth rate falls by around 0.14 percentage points, which is equivalent to an output loss of 5.4 percent over the twenty years since the reform. Ninety percent of the decline in growth is driven by incumbent innovation with the remainder working through a reduction in entry.

My quantitative results suggest that the pension reform overly tightened risk requirements on defined-benefit pension funds. The withdrawal of pension capital, in turn, not only led to lower investment by incumbent firms, but also had negative spillovers for prospective outside innovators through a *thinning out* of equity markets. While other investors demand partially compensated for pension schemes' divestment, the market response was not sufficiently elastic to fully offset the impact of the regulatory reform.

Literature. This paper adds to four strands of literature. First, I introduce a new framework linking equity markets and creative destruction.[1] Recent literature has focused on financial frictions in debt markets. In Aghion et al. (2025), we provide a model of endogenous financial frictions in a Klette and Kortum (2004) setting. See also the recent papers by Geelen et al. (2022) on leverage, Malamud and Zucchi (2019) on liquidity hoarding, Bustamante and Zucchi (2022) on the decline in real interest rates, and Akcigit et al. (2022b) on frictions in loan liquidation. The contribution of my paper is to introduce a framework

[1] A parallel literature studies risk premia and growth models with expanding varieties. These models predict a monotonically decreasing relationship between aggregate risk premia and growth (e.g. Kung and Schmid, 2015), while the Schumpeterian framework in this paper implies a non-linear relationship driven by the reallocation of resources towards outsiders whose innovation is more radical than incumbent firms'.

that connects segmented equity markets with growth. This opens the door to analysing how creative destruction and stock prices interact in general equilibrium, which could be useful for a series of applications connecting growth and finance: On the growth side, changes in risk premia are linked to firms' innovation incentives, making the model useful to study how equity market imperfections shape innovation incentives. On the finance side, the framework endogenises valuations as functions of innovation by incumbents and entrants, who generate endogenous tail risk, which makes it possible to explore how creative destruction affects asset prices (e.g. Kogan et al., 2017) and equity premia (e.g. Barro and Ursúa, 2012).

Second, I document that regulation-driven changes in investor demand that have persistent effects on stock prices can, in turn, impact firms' real investment decisions. As such, I contribute to the literature on the importance of quantities for asset prices (Gabaix and Koijen, 2021; Vayanos and Vila, 2021) and on the real effects of financial market segmentation. The empirical literature has focused primarily on demand shifts in bond rather than equity markets. Selgrad (2024) finds that an increase in demand for firms' bonds through central banks' quantitative easing operations can boost capital investment. Hubert de Fraisse (2024) documents that changes in government bond supply can trigger inflows into the corporate bond market and thus allow firms to scale up their long-term investment. Similarly, Coppola (2024) and Kubitza (2025) show how insurance companies' persistent demand for long-term bonds is associated with higher prices, lower capital costs, and higher investment rates at the firm level. Finally, Koijen et al. (2024) demonstrate that financial regulation can trigger large demand-driven flows that affect firms' cost of capital.

Third, I contribute new evidence on the impact of pension funds on firms' investment decisions. A series of papers has used index-inclusion instruments to show that institutional ownership is associated with higher innovation rates at the firm level (Aghion et al., 2013; Bena et al., 2017). There is also evidence that pension investment is linked to higher productivity (Beetsma et al., 2024), patenting (Pozzoli et al., 2024), and profitability (Harford et al., 2018). While this literature mostly relies on index inclusion instruments or event study evidence after a change in ownership composition, my paper uses a natural experiment to provide plausibly causal evidence on the impact of pension fund divestment on firm outcomes. An exception here is Giannetti and Laeven (2009) who use a pension reform in Sweden to show that increased pension fund stock market participation has a positive impact on valuations and corporate governance.[2] However, they do not study the effect on

[2] A common view within this literature is that stable investors, of which pension funds are one example, reduce funding uncertainty and encourage long-term investing (e.g. Derrien et al., 2013). The corporate finance literature refers to this as the *catering view* of ownership composition (e.g. Dong et al., 2012; Edmans, 2009; Edmans and Manso, 2011; Holmström and Tirole, 1993; von Thadden, 1995). Different types of shareholders alter managerial incentives and therefore investment strategies usually through investors' capacity to monitor.

firms' investment decisions, and none of the aforementioned papers analyse the impact on economic growth.[3]

Finally, this paper relates to the literature on the investment practices of public defined-benefit pension funds. While most of this literature has discussed the institutional details and pension scheme performance in the United States (e.g. Andonov et al., 2026; Andonov and Rauh, 2022; Andonov et al., 2012; Brown et al., 2015; Novy-Marx and Rauh, 2009, 2011) and the Netherlands (Jansen et al., 2023; Jansen, 2025), my paper focuses on the United Kingdom. The reform examined in this paper introduced strict guidelines for calculating pension schemes' funding status, requiring assets to be marked to market and eliminating adjustments for expected future scheme performance. As such, my results may be informative about the impact of institutional reform in other pension markets.

This paper proceeds as follows. In Section 2 I introduce the baseline model of risk aversion and economic growth, and derive the main predictions and comparative statics results. In Section 3 contains the empirical application. Section 4 uses the empirical findings to study the aggregate effects on economic growth. Section 5 concludes.

2 A Model of investor risk aversion and growth

Standard models of innovation-driven growth assume that investors are risk neutral and price firm equity at a risk-free rate. By construction, these models cannot speak to the impact of changes in investor composition or investor risk aversion on firm dynamics and growth. In this section, I introduce a model of endogenous growth with risk-averse investors. The starting point is a standard endogenous growth framework a la Aghion et al. (2014) and Acemoglu and Cao (2015) with innovation by incumbents and outsiders.

2.1 Environment

Time is continuous. Let $\{\Omega, \mathcal{F}, \mathcal{P}\}$ be a probability space with filtration $\mathcal{F}_t$ capturing the information at time $t \in \mathbb{R}_+$. There are four groups of agents: households, investors, outsiders, and incumbents, the latter comprising monopolistic intermediate producers and a competitive final good firm.

[3] Different to this literature, which concentrates on the effect of pension fund investment at the *micro* level, typically through firms' governance and incentive structure, in this paper I take a *macro* view and show how regulatory-driven changes in pension fund activity in equity markets can shape economic growth through the supply and price of funding. Relatedly, Scharfstein (2018) argues that the capitalisation rate of pension funds is positively related to development of capital markets across countries. Similarly, the rise in pension investment in private markets has been credited as one of the key drivers behind the venture capital boom of the 1980s (Kortum and Lerner, 2000; Gompers and Lerner, 2001).

2.1.1 Household

The representative household inelastically supplies L units of labour and saves in a risk-free bond. Its members discount the future at rate r and derive utility from consumption:

$$\mathrm{E}\left[\int_0^\infty \exp\left(-rt\right) dC_t \,\middle|\, \mathcal{F}_t\right]. \tag{2.1}$$

The linearity in (2.1) ensures that the household elastically supplies any quantity of the risk-free bond demanded by financial markets at the constant market clearing rate r.[4]

2.1.2 Production

The final good Y_t, the economy's numéraire, is produced in a two-stage process split between a competitive final good producer and a set of intermediate good monopolists.

Final good. The final good producer uses labour L_t^P and $i = 1, \ldots, N$ intermediate inputs, each of which comes at some quality level q_{it}. Higher-quality inputs have a higher marginal product. The production technology is represented by the nested Cobb-Douglas function:

$$Y_t = \frac{1}{1-\vartheta}\left(\sum_{i=1}^{N} q_{it}^{\vartheta} y_{it}^{1-\vartheta}\right)\left(L_t^P\right)^{\vartheta}, \tag{2.2}$$

where $1/\vartheta$ is the elasticity of substitution between different varieties of the intermediate good. With an input price $\varrho(q_{it})$ and wage w_t, the input demand functions are:

$$y_{it} = \varrho\left(q_{it}\right)^{-\frac{1}{\vartheta}} q_{it} L_t^P \qquad \text{and} \qquad L_t^P = \frac{\vartheta Y_t}{w_t}. \tag{2.3}$$

Intermediate goods. Intermediate producers make a static and a dynamic decision. The static decision is how much to produce for a given quality level q_{it}. A producer's technology turns $1-\vartheta$ units of the final good into one unit of an intermediate good at quality q_{it}. Taking demand (2.3) as given, the firm chooses its price $\varrho\left(q_{it}\right)$ to maximise profits $\pi(q_{it})$. The problem yields a linear profit function $\pi(q_{it}) = \pi q_{it}$, where $\pi \equiv \vartheta L_t^P$. The details are in Appendix A.1.2. The firm's dynamic decision involves determining how much it should invest to improve its product quality. I will discuss investment in innovation in Section 2.1.3.

[4] An equivalent interpretation is a small open economy where international investors provide capital at rate r but cannot hold domestic stocks. See Brunnermeier and Sannikov (2014, 2016) for a similar setup.

Aggregation. To aggregate the production side of the economy, I define the cumulative quality level of intermediate firms as:

$$Q_t \equiv \sum_{i=1}^{N} q_{it}. \tag{2.4}$$

Appendix A.1.2 shows that when summing across intermediate firms, output, production costs, wages, and profits grow with Q_t. I write firms' relative quality as $\varphi_{it} \equiv q_{it}/Q_t$.

2.1.3 Intermediate firms

Innovation. The quality of an intermediate input changes over time for three reasons. First, intermediate firms innovate. Investing $\zeta z_t^{\eta}\varphi_{it}$ units of labour increases the quality level at a flow rate z_{it}. Second, incumbents are at risk of creative destruction from outsiders at Poisson rate e_{it}. Whenever an outsider successfully innovates on a product line, quality jumps by $\nu > 1$ and the current incumbent is replaced. Third, firms are subject to aggregate risk. Combining these, product quality follows:

$$\frac{dq_{it}}{q_{it}} = z_{it}dt + \sigma_a d\mathcal{Z}_t + (\nu - 1)d\mathcal{J}_{it}, \tag{2.5}$$

where $d\mathcal{Z}_t$ is an aggregate risk factor with volatility σ_a and $d\mathcal{J}_{it}$ represents creative destruction risk at intensity $\mathrm{E}[d\mathcal{J}_{it}] = e_{it}dt$ on product line i. Investors demand a risk premium due to aggregate risk and creative destruction risk.

Stock returns. Firms' stocks are traded in a public market. Given (2.5), I denote the price of firm i's stock by $P(q_{it})$. The return consists of the dividend yield $[\pi(q_{it}) - \zeta z_{it}^{\eta}\varphi_{it}w_t]/P(q_{it})$ plus the capital gain and the creative destruction jump. Using Itô's lemma it follows that:

$$dR(q_{it}) = \mu_{it}dt + \varsigma_{a,it}d\mathcal{Z}_t - d\mathcal{J}_{it}, \tag{2.6}$$

with the short-hand coefficients for the drift and the volatility given by:

$$\mu_{it} \equiv \frac{\pi(q_{it}) - \zeta z_{it}^{\eta}\varphi_{it}w_t}{P(q_{it})} + \frac{P'(q_{it})z_{it}q_{it}}{P(q_{it})} + \frac{1}{2}\frac{P''(q_{it})\left(\sigma_a q_{it}\right)^2}{P(q_{it})} \quad \text{and} \quad \varsigma_{a,it} \equiv \frac{P'(q_{it})\sigma_a q_{it}}{P(q_{it})}. \tag{2.7}$$

Note that even though all firms load symmetrically on the aggregate risk factor, returns will be subject to different levels of volatility, $\varsigma_{a,it}$ because of the geometric Brownian motion assumption. The Appendix introduces heterogenous firms with different exposures to aggregate risk.

2.1.4 Investors

A unit mass of investors allocates their wealth between a risk-free bond, paying r, and N stocks with returns given by (2.7). A share β of these investors are *risk-constrained* investors, indexed by L, while the remaining $1-\beta$ are *market-driven* investors, indexed by M. One may think of the former as defined-benefit pension schemes and of the latter as retail investors. I will discuss their portfolio choice in turn.

Market-driven investors. Market-driven investors have mean-variance utility with risk aversion parameter $\tilde{\gamma}$ over instantaneous changes in their wealth level W_t^M. To ensure the model is consistent with balanced growth, I define the risk aversion parameter $\tilde{\gamma} \equiv \gamma/Q_t$ where Q_t is the aggregate quality level defined in (2.4).[5] Given portfolio holdings $\{\alpha_{it}^M\}_{i=1}^N$ for these stocks, investor wealth follows:

$$\frac{dW_t^M}{W_t^M} = rdt + \sum_{i=1}^{N} \alpha_{it}^M \left[(\mu_{it} - r)\, dt + \varsigma_{a,it} d\mathcal{Z}_t - d\mathcal{J}_{it} \right]. \tag{2.8}$$

Accordingly, each market-driven investor solves the following mean-variance optimisation problem by choosing the portfolio shares of equity:

$$\max_{\{\alpha_{it}^M\}_{i=1}^N} \; \mathrm{E}\left[dW_t^M\right] - \frac{\tilde{\gamma}}{2} \mathrm{Var}\left[dW_t^M\right] \qquad \text{s.t.} \qquad (2.8). \tag{2.9}$$

As shown in Appendix A.1.3, the solution to (2.9) is given by the demand function:

$$X_{it}^M = \frac{\mu_{it} - r - e_{it} - \tilde{\gamma}\varsigma_{a,it} \left(\sum_{j \neq i}^{N} X_{jt}^M \varsigma_{a,jt} \right)}{\tilde{\gamma}\left(e_{it} + \varsigma_{a,it}^2\right)}. \tag{2.10}$$

Here, $\mu_{it} - r - e_{it}$ is the excess return adjusted for the creative destruction risk that wipes out investors. Moreover, $\tilde{\gamma}\varsigma_{a,it}\left(\sum_{j\neq i}^N X_{jt}^M \varsigma_{a,jt}\right)$ corrects for the correlation between stocks due to aggregate risk. Finally, there are two source of volatility, idiosyncratic creative destruction shocks at e_{it} (the variance of a Poisson process), and aggregate shocks at $\varsigma_{a,it}^2$.

Risk-constrained investors. Risk-constrained investors' portfolio choice problem parallels their market-driven counterparts'. To that, I add one additional feature meant to capture regulation-driven changes in investors' de-facto risk aversion.

[5] This assumption ensures that investors' portfolio shares do not collapse to zero as their wealth level grows over time.

As before, investors also hold wealth W_t^C with its law of motion mirroring (2.8). Their risk aversion is $\tilde{\gamma}$. Different to market-driven investors, risk-constrained investors are subject to a penalty on large volatility in their wealth levels:

$$d\mathcal{P}\left(W_t^C; \tilde{\kappa}\right) = \tilde{\kappa} \operatorname{Var}\left[dW_t^C\right], \tag{2.11}$$

where $\tilde{\kappa} \geq 0$ parametrises the constraint's tightness. As with market-driven investors, I define the risk aversion parameter $\tilde{\gamma} \equiv \gamma / Q_t$ and $\tilde{\kappa} \equiv \kappa / Q_t$.[6] The penalty (2.11) can be interpreted as a value-at-risk constraint on investor net wealth limiting their exposure to volatile equity. In general, imposing a limit on portfolio volatility can be rationalised with an incentive problem, for example, a standard empire-building friction.

Risk-constrained investors have mean-variance preferences over instantaneous changes in their wealth level. They choose portfolio shares α_{it}^C for the $i = 1, \ldots, N$ stocks to solve:

$$\max_{\left\{\alpha_{it}^C\right\}_{i=1}^N} \mathrm{E}\left[dW_t^C\right] - \frac{\tilde{\gamma}}{2} \operatorname{Var}\left[dW_t^C\right] - d\mathcal{P}\left(W_t^C; \tilde{\kappa}\right) \qquad \text{s.t.} \qquad (2.11). \tag{2.12}$$

As shown in Appendix A.1.3, the optimisation problem (2.12) yields a demand function for each firm i's stock of:

$$X_{it}^C = \frac{\mu_{it} - r - e_{it} - (2\tilde{\kappa} + \tilde{\gamma})\varsigma_{a,it}\left(\sum_{j \neq i}^N X_{jt}^C \varsigma_{a,jt}\right)}{(2\tilde{\kappa} + \tilde{\gamma})\left[e_{it} + \varsigma_{a,it}^2\right]}. \tag{2.13}$$

When $\tilde{\kappa} = 0$, the solution in (2.13) coincides with the portfolio share for market-driven investors in (2.10). A larger penalty reduces the share of wealth invested in stocks.

2.1.5 Stock price and investment

Firms invest in innovation to maximise their stock price. There are no incentive frictions related to the choice of z_{it}, which may produce a misalignment in incentives between stockholders and the firm's management.[7] For the model to be consistent with balanced growth, I assume that each firm has Q_t units of its stock outstanding, so the aggregate supply remains constant relative to the size of the economy.

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[6] Whether market-driven investors and risk-constrained investors have the same coefficient of risk aversion γ does not affect the qualitative or quantitative model predictions as one can always rescale the parameter κ.

[7] One rationalisation of this assumption is that the firm's management cares about the stock price, for example because compensation is tied to performance via stock options or incentive pay. A literature starting from Stein (1989) studies the feedback between innovation and the stock market. See also Aghion and Stein (2008) and Narayanan (1985). As my model does not have frictions on the firm side, maximising the stock price is equivalent to maximising the expected instantaneous return (2.6) for the average investor.

In equilibrium, investor risk aversion and the volatility of the firm's quality process give rise to a risk premium on its stock. As shown in Appendix B.1.4, market clearing in the stock market implies that the firm's Hamilton-Jacobi-Bellman equation for the stock price is given by the following expression:

$$\max_{z_{it}} \left\{ \begin{array}{c} \pi(q_{it}) - \zeta z_{it}^{\eta} \varphi_{it} w_t + P'(q_{it}) z_{it} q_{it} + \frac{1}{2} P''(q_{it}) \left(\varsigma_{a,it} q_{it}\right)^2 \\ -\lambda \varsigma_{a,it} \mathcal{B}_{it} P(q_{it}) - \left[r + \lambda \varsigma_{a,it}^2 + (1+\lambda) e_{it}\right] P(q_{it}) \end{array} \right\} = 0, \tag{2.14}$$

where

$$\lambda \equiv \frac{\gamma(2\kappa + \gamma)}{2\kappa(1-\beta) + \gamma} \qquad \text{and} \qquad \mathcal{B}_{it} = \beta \sum_{j \neq i}^{N} X_{jt}^{C} \varsigma_{a,jt} + (1-\beta) \sum_{j \neq i}^{N} X_{jt}^{M} \varsigma_{a,jt}. \tag{2.15}$$

The parameter λ captures average market risk aversion. It is a weighted average of effective investor risk aversion γ and $\gamma + 2\kappa$, where the weights are the relative shares of investors β and $1 - \beta$ in the market. Note that this expression is independent of Q_t as the number of stocks issued by the firm grows over time such that the relative supply of stocks grows with the size of the economy. When $\gamma = \kappa = 0$, investors are risk-neutral and (2.14) collapses to the standard expression known from other endogenous growth models.

The term $\mathcal{B}_{it}$ captures by how much the firm's valuation is reduced due to its correlation with other firms in the market. Because of CARA demand, this factor depends on investors' positions. The first line of the firm's value function (2.14) contains the dividend and the changes in the stock price due to innovation and quality fluctuations. The second line is the discounted value of the stock, including a two-part risk premium, due to aggregate risk and creative destruction.

The first-order condition associated with the problem in (2.14) equates the marginal cost of choosing a higher drift, both in terms of the physical investment cost and the financial cost of a higher risk premium, to the resulting change in the stock price:

$$\zeta \eta z_{it}^{\eta-1} \varphi_{it} w_t = P'(q_{it}) q_{it}. \tag{2.16}$$

2.1.6 Entry, growth, and the firm size distribution

Entry. Creative destruction comes from a large number of prospective outsiders. Entry is directed. Outsiders can invest $\psi(e_{it})\varphi_{it}$ units of labour to innovate on a product line $i = 1, \ldots, N$ with intensity e_{it}. The dependence on $\varphi_{it} = q_{it}/Q_t$ reflects that it is easier to innovate on product lines that are further away from the frontier and have lower quality.

A successful innovation attempt introduces a technology that allows the outsider to produce on line i with quality νq_{it}, which exceeds the incumbent's quality q_{it} by a step size $\nu > 1$. The outsider dislodges the incumbent and becomes the new monopolist. After successful entry, the new firm issues Q_t units at price $P(\nu q_{it})$. The free-entry condition is:

$$e_{it}\,\mathrm{E}\left[P(\nu q_{it})\right] = \psi\left(e_{it}\right)\varphi_{it} w_t. \tag{2.17}$$

Growth. The expected growth rate of aggregate product quality is $\mathrm{E}\left[dQ_t/Q_t\right] \equiv g_t dt$. The change in the aggregate quality level itself is $dQ_t/Q_t = \sum_{i=1}^{N}(dq_{it}/q_{it})\varphi_{it}$. Thus, taking expectations, and using (2.5) for dq_{it}, as well as noting that $\mathrm{E}\left[d\mathcal{J}_{it}\right] = e_{it}dt$, yields:

$$g_t = \sum_{i=1}^{N}\left[z_{it} + (\nu - 1)e_{it}\right]\varphi_{it}. \tag{2.18}$$

The firm size distribution is characterised by a Kolmogorov-forward equation, which can be found in Appendix A.1.6.

2.1.7 Market clearing

There are three plus $2 \times N$ markets in the economy: one each for the final good, labour and the risk-free bond, as well as N markets for intermediate goods and the same number for firms' stock. The omitted market clearing conditions are listed in Appendix A.1.5.

Final good, intermediate goods, production labour, and bonds. The markets for the final good and the bond clear because households, whose linear utility allows for consumption to be positive or negative, are willing to borrow and lend arbitrary amounts at the risk-free rate r, equal to their discount rate. Intermediate good markets clear by the first condition in (A.2).

Labour. Households inelastically supply L units of labour. $\vartheta Y_t/w_t$ units are used in final good production as specified in (2.3), ψe_{it} units as an input in entry, and $\zeta z_{it}^{\eta}\varphi_{it}$ for incumbent innovation by each of the N monopolists. The market clearing condition is:

$$L = \frac{\vartheta Y_t}{w_t} + \sum_{i=1}^{N}\left[\psi\left(e_{it}\right) + \zeta z_{it}^{\eta}\right]\varphi_{it}. \tag{2.19}$$

Stocks. As for stock market clearing, each firm inelastically supplies Q_t units of stock such that the total amount of stocks outstanding grows with the size of the economy. Demand comes from $1-\beta$ market-driven investors and β risk-constrained investors according to (2.10) and (2.13). The market clearing condition for stocks $i = 1, \ldots, N$ is:

$$Q_t = (1-\beta)X_{it}^M + \beta X_{it}^C. \tag{2.20}$$

2.2 Equilibrium

We can now discuss the equilibrium in this economy. I will first formally define the growth path and then derive firms' equilibrium innovation.

2.2.1 Equilibrium definition

Definition 1. *The economy is on a stochastic balanced growth path if all aggregate variables – output Y_t, consumption C_t, profits Π_t, wages w_t, and intermediate production costs Y_t^P – grow at a common expected growth rate g_t, the growth rate of aggregate quality Q_t defined in* (2.18)*; the risk-free rate is constant at r, the risk-neutral household's discount rate defined in* (2.1)*; and stock returns follow the process in* (2.6)*. Agents' optimisation problems satisfy the following conditions:*

(i) households choose consumption and savings according to (2.1)*;*

(ii) the final good firm makes zero profits. Its demand functions for $i = 1, \ldots, N$ intermediate inputs and production labour are given by (2.3)*;*

(iii) intermediate firms set prices and quantities, and earn profits according to (A.2)*; they maximise their stock price* (2.14) *by choosing the drift of* (2.5) *according to the first-order conditions* (2.16)*;*

(iv) market-driven investors maximise (2.9) *subject to the law of motion* (2.8) *by choosing their portfolio share of each risky stock $i = 1, \ldots, N$ according to* (2.10)*;*

(v) risk-constrained investors maximise (2.12) *subject to a law of motion equivalent to* (2.8) *and* (2.11) *by choosing the portfolio share of stocks $i = 1, \ldots, N$ according to* (2.13)*;*

(vi) product quality follows the Kolmogorov-forward equation (A.14)*;*

(vii) and the free entry condition (2.17) *holds;*

such that the markets for the final good, labour, the risk-free bond, N intermediate goods, and N stocks described by (A.9), (A.10), (A.11), (A.12), *and* (A.13) *clear.*

2.2.2 Solving for equilibrium

From now on, I drop the subscripts to simplify the notation and indicate that the solution is symmetric across firms. I guess that the firm's stock price is linear in its quality level, $P(q) = Pq$. Equation (2.7) implies that the volatility of the quality process is $\varsigma_a = \sigma_a$. The price equation (2.14), the firm's first-order condition (2.16), the free-entry condition (2.17), and the labour market clearing condition (2.19) define a system of four equations in four unknowns, the stock price P, the innovation rate z, the entry rate e and the amount of labour used in the production of the final good L^P. The growth rate follows once entry and incumbent innovation have been determined. The analytical solution is described in Proposition 1 and the step-by-step derivation can be found in Appendix A.2.

Proposition 1. *Consider the equilibrium of the model described in Definition 1. Incumbent investment and entry rates are given by:*

$$z = \left[\frac{(1-\vartheta)P}{\zeta\eta\vartheta}\right]^{\frac{1}{\eta-1}} \qquad \text{and} \qquad \psi\left(e\right) = \frac{\nu(1-\vartheta)eP}{\vartheta}. \tag{2.21}$$

The equilibrium growth rate is:

$$g = (\nu - 1)z + e, \tag{2.22}$$

and the firm's stock price and risk premium are

$$P = \frac{\pi - \zeta z^{\eta}\vartheta/(1-\vartheta)}{\mu + z} \qquad \text{and} \qquad \mu - r = (1+\lambda)e + \lambda N\sigma_a^2. \tag{2.23}$$

Finally, the amount of labour used in final good production is pinned down by $L^P = L - \psi(e) - \zeta z^{\eta}$, final output is $Y = \frac{1}{1-\vartheta}L^P Q$, and the equilibrium wage is $w = \vartheta/(1-\vartheta)Q$.

Proof. Appendix A.2. □

2.3 Comparative statics

I will now discuss the comparative statics of a change in risk-constrained investors' risk-taking capacity on firm's investment decisions and on economic growth. I will start with the effects on incumbent investment and outside innovation in partial equilibrium and then move on to the overall effect on economic growth.

Firm-level effect. Starting with the impact at the firm level, a tighter regulatory constraint κ reduces risk-constrained investors' demand for stocks (2.13). This drives up the market risk premium (2.23). Holding fixed the intensity of outside innovation e, we have:

$$\frac{\partial \mu}{\partial \kappa} = \left(e + N\sigma_a^2\right)\frac{\partial \lambda}{\partial \kappa} > 0. \tag{2.24}$$

A higher risk premium pushes down stock prices P in (2.23) and therefore lowers incumbent firms' incentive to invest in innovation, as can be seen in (2.21). Firms cut back on investment, which further reduces the stock price. The proposition below summarises these results.

Proposition 2 (Incumbent innovation effect). *A tightening of financial regulation on a subset of investors raises the market risk premium, reduces stock prices, and investment by incumbent firms in partial equilibrium:*

$$\frac{\partial \mu}{\partial \kappa} > 0, \qquad \frac{\partial P}{\partial \kappa} < 0, \qquad \textit{and} \qquad \frac{\partial z}{\partial \kappa} < 0. \tag{2.25}$$

Entry effect. An tightening of financial regulation on a subgroup of investors, as captured by an increase in κ, affects innovation by outsiders through two distinct channels. First, a higher risk premium on incumbent investment lowers their stock prices on impact, thus decreasing the expected value of becoming an incumbent firm in the future. This negative *risk premium effect* discourages entry.

Second, in response to a higher risk premium, incumbent firms scale back their investment as described above. This reduces their demand for labour and therefore the equilibrium wage, resulting in lower innovation costs and therefore a reallocation of resources towards production and outsiders. This standard *reallocation effect* can be found in most endogenous growth models (Aghion et al., 2014, 2025) and counteracts the negative risk premium effect on outside innovation.

Differentiating the free-entry condition with respect to κ illustrates these two counteracting effects:

$$\underbrace{\frac{\nu\vartheta}{1-\vartheta}\cdot\frac{dP}{d\kappa}}_{\text{Risk premium effect}} - \underbrace{\frac{\partial}{\partial \kappa}\left[\frac{\psi(e)}{e}\right]}_{\text{Reallocation effect}} \lesseqgtr 0. \tag{2.26}$$

Which effect dominates in equilibrium crucially depends on how much small changes in the cost of outside innovation affect the probability of successful innovation. A sufficient statistic for the net effect on entry is the expression $\psi(e)/e$, which captures the R&D efficiency of entrant firms, that is, the flow of innovation e per unit cost $\psi(e)$.

When small variations in the available amount of labour result in large changes in the probability of successful outside innovation, the reallocation effect can be strong and lead to a net increase in entry, compensating for the drop in stock prices due to higher risk premia. Put differently, when entry costs are highly convex, the reallocation effect is weak and entry falls in response to an increase in investor risk aversion. Proposition 3 summarises.

Proposition 3 (Entry effect). *The net effect of a tighter regulatory constraint κ on entry innovation is ambiguous and depends on the elasticity of entry cost $\psi(e)/e$. Net entry is positive if and only if $\nu P > \psi'(e)\frac{\vartheta}{1-\vartheta}$.*

Proof. Appendix A.3. □

Aggregate effect. The expected growth rate is a weighted average of incumbent innovation z and entry innovation e. A tightening of the regulatory constraint κ therefore affects growth via same three effects on the risk premium, reallocation between incumbents and outsiders, and general equilibrium amplification. One can distinguish two cases.

When the reallocation effect is weak, entry rates and incumbent innovation are decreasing in κ, and so is growth. When the reallocation effect is sufficiently strong, however, entry rates are increasing in κ. In this case, more innovation by outsiders can compensate for lower incumbent investment and the relationship between κ and the growth rate is hump-shaped. For low degrees of regulation, the growth rate is increasing in the tightness of the regulatory constraint on risk-constrained investors. When the constraint becomes *too tight*, growth falls. As a consequence, there must also be a growth-maximising level of regulation κ^* that balances the positive entry response against lower incumbent innovation. Proposition 4 summarises.

Proposition 4 (Growth effect). *Assume the reallocation effect is not sufficiently strong to offset the risk premium effect. Then, the growth rate is decreasing in the tightness of the regulatory constraint κ.*

Proof. Follows directly from Propositions 2 and 3. □

2.4 Simple case

There are three simple cases that allow for full characterisation in closed form and are useful to illustrate the general equilibrium effects of outside innovation. See Appendix A.4 for the details of each case.

First, almost all standard models of firm dynamics work with the knife-edge case of linear entry costs $\psi(e) = \psi e$ (e.g. Aghion et al., 2014; Akcigit and Kerr, 2018; Klette and

Kortum, 2004). In that case changes in investor risk aversion have no impact on aggregate innovation. The intuition is as follows. With linear entry cost, the stock price is fully pinned down by the wage rate $\nu P = \psi\vartheta/(1-\vartheta)$. In that case, any change in incumbent innovation is fully offset by a fall in firm entry, which in turn raises the value of the firm to offset the rise in the risk premium. Incumbent innovation is $z = [\psi/(\zeta\eta\nu)]^{1/(\eta-1)}$ and does not depend on investor risk aversion. Put differently, an increase in investor risk aversion depresses entry exactly to offset any impact on prices.

Second, a more interesting case that allows for full characterisation in closed form arises when entry costs are convex (e.g. Acemoglu et al., 2018; Berlingieri et al., 2025). A simple example is the quadratic entry cost function $\psi(e) = e^2/2$. Given that a change in the tightness of regulation κ is isomorphic to a change in market risk aversion λ, the comparative statics are:

$$\frac{dP}{d\lambda} = -\frac{\left(e + N\sigma_a^2\right)P}{\Lambda}, \qquad \frac{de}{d\lambda} = -\frac{e}{P}\cdot\frac{dP}{d\lambda} \qquad \text{and} \qquad \frac{dz}{d\kappa} = -\frac{z}{(\eta-1)\,P}\cdot\frac{dP}{d\lambda}, \tag{2.27}$$

where $\Lambda \equiv \mu - \left(\frac{\eta+\vartheta}{\eta-1}\right)z + (1+\lambda)\,e + 2\,(1-\vartheta)\,\nu e > 0$ is the firm's effective discount factor. Entry costs scale more than one-to-one with the rate of outside innovation e, such that the reallocation effect cannot offset the decrease in the present value of entry (the stock price) due to the risk premium effect.

Third, with concave entry costs, for example $\psi(e) = \psi \ln e$, the cost of entry innovation increases at a slow enough pace such that the reallocation effect towards outside innovators can compensate for the fall in stock prices. Outside innovation increases as a result. This has a knock-on effect on incumbent innovation, as additional outside innovation raises the risk of creative destruction, lowers the stock price, and therefore reduces incumbents' willingness to invest, knowing they are more likely to be replaced in the future.

2.5 Discussion

This section discusses the key assumptions in the model and how to interpret the role of pension funds in this context.

The hump-shape and its drivers. As Propositions 3 and 4 illustrate, the net effect of a change in effective market risk aversion, such as the pension fund reform studied in the empirical part of the paper, on economic growth is not necessarily negative. In particular, to the extent that tighter constraints on equity investment allow for reallocation of resources between incumbent and outside firms, any negative impact on the innovation rate of affected firms may be dampened or even overturned in general equilibrium.[8] This mechanism is driven by two assumptions.

First, in the model, reallocation can lead to an increase in growth because there is a gap in innovation impact between incumbents and outsiders. Outsider innovation boosts product quality by a net step size $\nu - 1$, while incumbent innovation advances product quality only incrementally. When incumbents and outsiders are equally effective, an increase in the risk premium driven by tighter regulation on investors always reduces growth. The empirical evidence suggests that entry innovation indeed tends to be more radical than incumbent innovation (Akcigit and Kerr, 2018). In fact, much of the discussion about the role of financing, and in particular venture capital, rests on the insight that innovation by outsiders is an important dimension of radical technical change (Akcigit et al., 2022a).

Second, the entry response in the model is only positive because the tightening in regulation on risk-constrained investors disproportionately affects the cost of capital for incumbent firms. In the baseline model, I focus on the case when outside innovation is only financed by inside equity, while incumbent investment depends on equity markets. This assumption broadly reflects the fact that smaller firms receive relatively less capital from larger institutional investors such as pension funds (Robb and Robinson, 2014). In a model in which outside innovators received the same fraction of their financing from pension funds and also shared the same risk profile, the increase in the risk premium due to regulatory tightening would equally reduce incumbent and outsider innovation. Any model extension in which outsiders are funded in stock markets would strengthen the negative effect on growth coming from tighter regulation. When outsider innovation is riskier than incumbent innovation, a rise in effective risk aversion would still lower growth, and the reallocation channel would need to be particularly strong to generate a positive entry response.

Inside versus outside innovation. One may wonder about the distinction between incumbents and outsiders in the context of the model. In the standard Schumpeterian growth framework, entrants are typically interpreted as small start-ups or entrepreneurial innovators who invest in innovation to unseat larger incumbents. An alternative interpretation in the context of my model is of incumbents as large public companies who make up the market index and of outsiders as the segment of firms just below that. This interpretation is convenient because a large fraction of institutional investment tends to be concentrated in those large indices that serve as their investment benchmark. As such the reallocation chan-

[8]This trade-off also distinguishes the Schumpeterian framework from Romer (1990) or other AK-type endogenous growth models, in which higher risk premia unambiguously reduce the quantity and quality of innovation.

nel uncovered in the model could be thought of as a reallocation not between incumbents and market entrants, but between larger public and smaller private firms.

The model therefore also yields additional predictions on the impact of regulation on *market concentration*. One interpretation of incumbent firms in the baseline model is that of large, listed firms. Although there has recently been a move towards private markets, pension funds and other institutional investors invest primarily in these large public firms. Entrant firms in the model can be interpreted as outside innovators, who are not necessarily small, but who have higher cost of capital, for example because they do not form part of an investment index. Any policy change that affects risk-constrained investors' demand for equities changes the cost of capital for incumbents and, through their investment policies, factor prices in secondary markets, which crowds out entry. In the model, crowding out happens through the labour market, but it could well be through other markets with inelastic supply of factor inputs, for instance machines or skilled labour.

The model therefore links institutional investment, market concentration, and growth, and could be used in future work to provide a financial explanation for the rise of superstar firms (Autor et al., 2020). Jiang et al. (2025), for instance, provide a theory linking passive investment and market concentration. In Appendix B.3, I extend the model to allow for heterogeneous firms and show that a decrease in risk-constrained investors' equity demand decreases market concentration and reduces the average firm size.

Exogenous risk-free rate. The assumption that households are risk neutral in (2.1) leads to a constant risk-free rate r. This assumption is made to ensure that households absorb any demand or supply of bonds coming from risk-averse investors. As discussed at the top, one might think of a small open economy where households proxy for international investors who access the stock market via local (risk-averse) intermediaries (Brunnermeier and Sannikov, 2014).

Of course, one might be concerned that changes in the economy's growth rate interact with the risk-free rate. While standard endogenous growth models typically do not assume linear household utility, these models neutralise any interest rate effects via log-utility. That is, the interest rate always adjusts to accommodate the chosen growth rate in equilibrium.

In the case of more general CARA utility, a reduction in the growth rate can lead to a reduction in the real interest rate and therefore amplification of any growth effects in general equilibrium. Because firms' innovation choices only depend on the risk-free rate via firm values P, any amplification via changes in the real rate r would amount to a third-order effect on growth, leaving the direction of the main comparative statics results unaffected.

3 Empirical evidence

To empirically test the mechanism proposed in Section 2, I draw on a regulatory change in the British defined-benefit pension sector, which led to an increase in de-facto risk aversion among pension schemes and a large and persistent divestment from equity markets. Section 3.1 introduces the institutional background. Section 3.2 discusses the aggregate consequences of the reform before Section 3.3 provides evidence on the mechanism with a series of firm-level regressions.

3.1 Data and institutional background

3.1.1 Data

Measuring the impact of pension investment on non-financial firms' outcomes requires a dataset of pension fund stock holdings linked to firms' balance sheet and earnings information. Unfortunately, the availability of micro data on pension funds in Britain has been historically very limited. The pension fund holdings in this paper comes from annual reports, accounts, actuarial valuation reports, and other financial data obtained from a variety of archival sources. I supplement this data with holdings data for pension funds from Morningstar, and Refinitiv. Where available, I consolidate mutual fund holdings data using Factset. In total, the dataset contains holdings for 98 British defined-benefit pension funds.

I match these holdings data to firm information using Compustat Global. I restrict the sample to non-financial firms headquarter in the UK and supplement the dataset with stock price data from CRSP. The final dataset is an unbalanced panel of 98 pension funds' stock holdings data matched to non-financial firms' balance sheet, income statement, and stock price data covering the financial years 1999 to 2023. Summary statistics and details on the pension fund data can be found in Appendix C.

3.1.2 Institutional background

The reform. In March 2001, after a series of reform attempts in the 1990s, the government announced its intention to overhaul the regulation of defined-benefit pension schemes. A consultation process culminated in the Pensions Act 2004, which strengthened regulatory oversight and expanded the powers of *The Pensions Regulator* (TPR).

The most important change was the introduction of the *Statutory Funding Objective* (SFO) for defined-benefit pension schemes in Section 222 of the Act. Under the SFO, the ratio of a pension scheme's assets to the discounted value of its future pension obligations would serve as the indicator of a scheme's financial health. If this *funding level*, formally defined as:

$$\mathcal{F}_t = \frac{\text{Market Value of Assets}}{\text{Present Value of Future Pension Obligations}}, \tag{3.1}$$

would take a value less than one, a pension scheme was to be treated as *underfunded*. That is, it would not be in a position to cover its obligations were they to fall due today.[9]

Calculating the funding level (3.1) requires two assumptions that were not provided for in the Act: first, whether the market environment should be factored into the calculation of asset values; and second, the discount rate to be applied to future pension obligations. Under the provisions of the Act, the authority to determine the appropriate valuation methodology was delegated to TPR. The details were to be set out in a regulatory directive. The provisions took effect on 1 April 2005 with the start of the financial year 2005/2006.

The TPR directive. In June 2005, the regulator published its directive on the interpretation and enforcement of the SFO (The Pensions Regulator, 2005). In the document, TPR clarified how it would evaluate pension funds' financial position as defined in (3.1):

1. When determining the market value of assets, only the current actuarial valuation of assets would be used. Current asset composition and investment strategies would *not* be taken into account when measuring fund deficits and mitigating factors such as recent equity market under-performance would not favourably affect valuations.

2. The relevant discount rate to calculate the present value of future obligations would be tied to the yield on long-dated, index-linked gilts but would not take into account expected returns on equity investments.[10]

3. Pension funds that were seen to be in bad financial health, for example due to a quick deterioration of their funding level, would be subject to regulatory intervention. Even for funds that were projected to close their deficits, the existence of a long recovery period itself would be seen as a trigger for regulatory intervention. As a measure of

[9]The motivation for the reform was to gain an accurate picture of funding levels in the defined-benefit sector at current market prices, which would help identify systemic deficits and root out risk-taking in the industry seen by many as excessive (e.g. Myners, 2001). In contrast, Giesecke and Rauh (2023) discuss how U.S. DB pension schemes base their discount rates on expected returns, which naturally implies higher funding levels relative to risk-free rates and may belie systematic deficits due to risky investment practices.

[10]To be precise, TPR outlined that it would use the so-called *Section 179 Deficit* (PPF Levy) rather than the accounting deficit under FRS17/IAS19 for its assessment of scheme funding status. FRS17/IAS19 deficits are based on AA yields to discount liabilities and take into account estimates for salary, inflation and future equity returns. Contrary to that, the Section 179 Deficit approach uses the market value of fund assets and discounts future liabilities at the relevant index-linked gilt rate at -50 basis points. The deficit is then calculated as the ratio of assets over the net-present value of liabilities.

last resort, the regulator reserved the right to close down schemes and transfer their assets into a master fund.

Timing of events. The government first announced its intention to reform defined-benefit pension schemes in 2001 with the publication of the Myners report (Myners, 2001). The reform proposal was first circulated as a government *White Paper* on 11 June 2003 (Department for Work and Pensions, 2003). Draft legislation was introduced as the *Pensions Bill* by the Pensions Secretary on 10 February 2004 (Lourie et al., 2004). The Bill was passed by both Houses over the summer and received Royal Assent on 18 November that year, thus becoming the Pensions Act 2004. The provisions took effect on 1 April 2005, and the regulator's directive was published in June of that year. Throughout this paper, I will treat the financial year 2002-2003, which ended on 31 March 2003, as the baseline pre-reform year.

3.2 Aggregate consequences

Effects on investment strategies. The regulator's interpretation of the SFO and the proposed methodology had an immediate effect on DB funds' investment strategies. Three aspects are particularly important. First, marking equity positions to market without accounting for macroeconomic conditions implied that DB pension funds with large equity position would experience fluctuations in their funding levels due to short-run movements in stock prices. As discount rates would not be adjusted for expected returns, the liabilities of an all-equity fund would be discounted at the same rate as those of an all-bond fund. These changes introduced substantial regulatory risk for funds with large equity positions.

Second, because the discount rate was tied to the yield at the long end of the gilts market, any movement in long-term rates would directly affect the present value of future pension obligations (3.1). Funds with a duration mismatch between assets and liabilities became exposed to mechanical fluctuations in their funding levels. Buying long-dated gilts or the fixed leg of an interest rate swap could be used to hedge that duration risk.

Third, the SFO came into effect during a period of low equity valuations and falling interest rates. In early 2004 the yield on the 30-year gilt stood at just under 4 percent, less than half its 1998 level. This timing coincidence made the first two issues even more pressing. Because of their losses in equity markets, in particular between the financial years 2001 and 2002, most schemes were underfunded under the SFO.

The directive was met with apprehension in the industry. At the time, it was variously described as a "*shot across the bows*" (Cohen, 2006) and "*exceedingly bearish*" for pension schemes (Barclays Capital Research, 2006). As I will show in the next section, pension funds

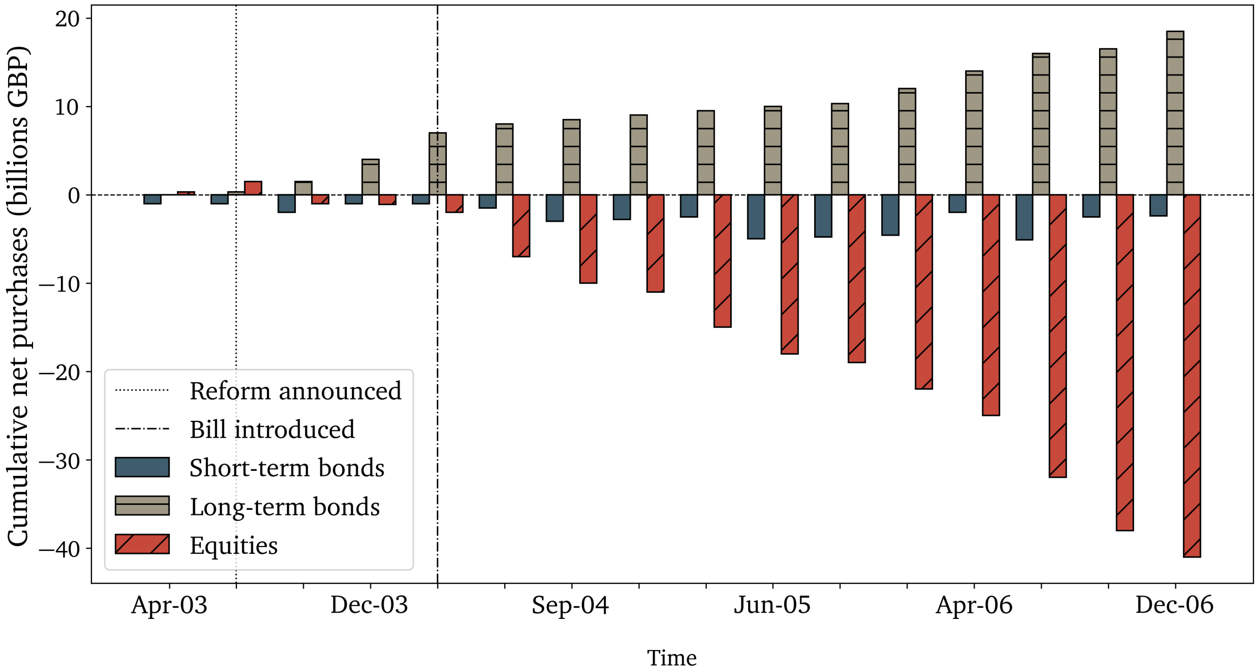

Figure 3.1: Pension fund cumulative net purchases, 2003-2006.

Flows are shown at market prices and are cumulated across calendar months. Red bars refer to equities, blue and beige bars to bonds. The omitted category is interest rate derivatives.

reacted by selling equities and buying long-dated gilts. My identification strategy in Section 3.3 will exploit that pension schemes with lower funding levels prior to the reform were more affected by the funding requirements and therefore sold larger quantities of stocks.

Withdrawal from equity markets. Figure 3.1 shows pension schemes' cumulative purchases by asset class from 2003 to 2006. Over this period, pension funds sold £40bn in equities and shifted towards gilts and derivatives, predominantly interest rate swaps at long maturities (Greenwood and Vayanos, 2010; Islam, 2007). Equity sales rose to £65bn by mid-2007, see Appendix D.2. The timing of the sales aligns with the passage of the pensions bill, which then accelerated with the release of the regulatory directive in 2005. Appendix D.1.1 documents that during this period the average portfolio share of domestic equities on schemes' balance sheets fell from 47 percent in April 2002 to 36 percent in April 2007.[11]

Equity sales during this period were limited to pension funds. Other large institutional investors, such as life insurance companies, did not reduce their exposure to equities. As insurers and DB pension funds both face long-dated liabilities, both investor types are sensitive to the funding risk due to movements in long-term interest rates. As I show in Appendix

[11]Appendix D.1.2 shows that the decline was driven by quantities rather than prices. If equity valuations had remained stable, the decline in the equity portfolio share would have been even larger than observed.

D.2, insurance companies increased their exposure to domestic equities during that period, while a shift into fixed income markets did not occur.

The absence of equity sell-offs among large insurance companies supports the view that the patterns in the DB sector were driven by the pensions reform rather than a change in macroeconomic conditions, which happened to coincide with the announcement or implementation of the reform. This view is also reflected in the investment reports that major banks were circulating at the time (e.g. Barclays Capital Research, 2005, 2006; Islam, 2007).

The importance of regulatory risk. Why did pension funds start to sell domestic equities after the reform? Under the new SFO, holding large quantities of equities would expose pension schemes to large fluctuations in their funding ratios. In fact, the general perception at the time was that domestic equities in particular exposed pension funds to too much volatility in their funding levels under the SFO relative to expected returns (Barclays Capital Research, 2006).

Although there was considerable variance in investment strategies prior to the reform, the shift out of equities in response to the reform occurred across pension funds, and so one may expect that other factors played a role in the aggregate divestment. However, if the sell-off was driven by changes in risk-taking capacity, one would expect schemes with lower funding levels prior to the reform, who would be more likely to be investigated under the new regulation, to decrease their equity holdings by more than schemes with relatively higher funding levels.

A particular feature of a previous reform, the Pensions Act 1995, allows me to test this hypothesis directly. The Act required defined-benefit pension funds to commission a triennial fund valuation with an accredited actuary. These valuation reports contain a detailed projection of future pension obligations and an official estimation of each scheme's funding level. The latest valuation cycle prior to the announcement of the Pensions Act 2004 was completed in April 2002. To test whether the pre-reform funding status is indeed predictive of the following divestment from equity, I estimate the following regression:

$$\Delta\left(\text{UK Equity Portfolio Share}\right)_{i,2007-2003} = \alpha + \beta \mathcal{F}_{i,2002} + \varepsilon_i, \tag{3.2}$$

at the fund level. Here, $\mathcal{F}_{i,2002}$ is a pension scheme's funding level in 2002 as defined in (3.1). The results of regression (3.2), as reported in Table D.1 in Appendix D.3.1, confirm this hypothesis: Pension schemes that had a greater shortfall in assets compared to their discounted liabilities pre-reform reduced their equity portfolio share by more. On average, a one percentage point lower funding ratio in 2002 corresponds to a 0.2 percentage point

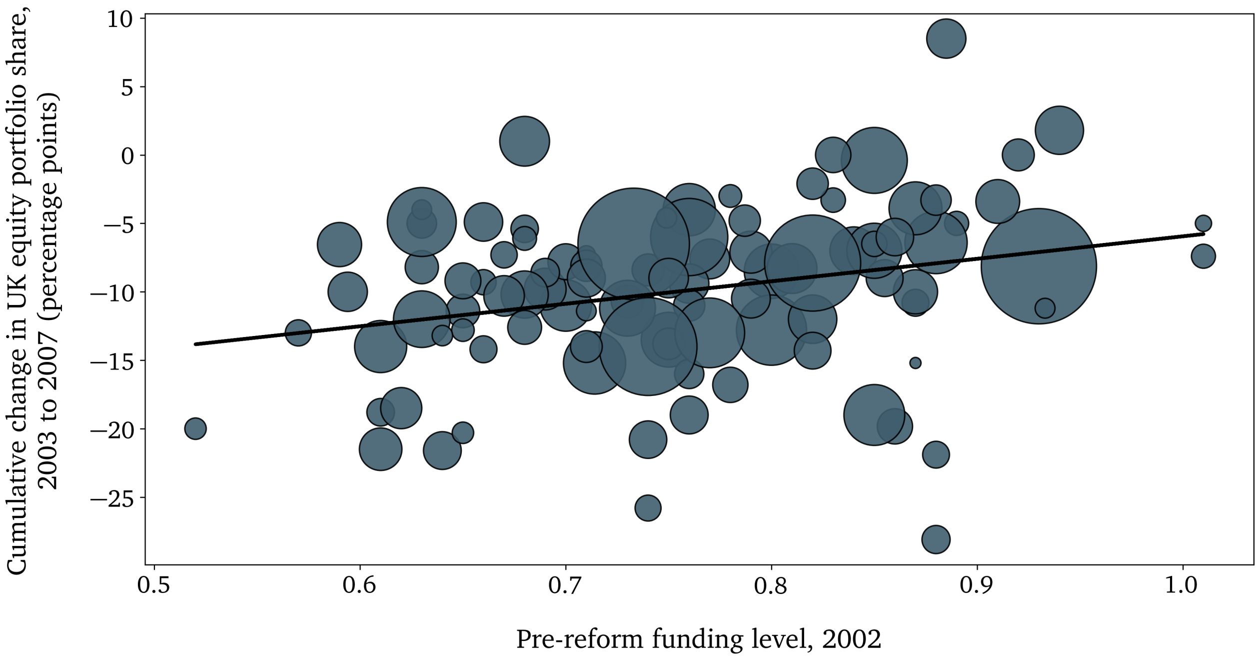

Figure 3.2: Pension fund changes in U.K. equity share.
Bubble sizes capture fund size by net assets in 2002. The line of best fit comes from regression (3.2).

further decrease in the equity share between 2003 and 2007. The shift out of equities was therefore particularly concentrated in pension schemes that were underfunded prior to the reform as defined under the SFO. Figure 3.2 contains a plot of the change in the U.K. equity portfolio shares against pre-reform funding levels.[12]

This differential shift out of equities based on schemes' pre-reform funding level is the main source of exogenous variation in my empirical analysis that follows in the next section: *Firms with a higher proportion of their stock held by underfunded pension schemes before the reform saw a larger sell-off of their stocks after the reform compared to other firms*. I will describe the precise empirical specification in the next section.

3.3 The effect on firm-level investment

This part of the paper contains the main empirical results. In Section 3.3.1, I discuss the specification for my firm-level regressions. Section 3.3.2 presents the empirical results. Section 3.3.3 discusses potential threats to identification and provides robustness exercises.

[12] To show that this pattern is not driven by changes in equity prices, I repeat the analysis in Figure 3.2 with holdings normalised to 2003 prices. See column (2) of Table D.1 in Appendix D.3.1.

3.3.1 Main specification

The baseline specification is a shift-share approach, using the pre-reform firm-level share of equity held by pension funds as the *share* and subsequent fund-level equity sales as the *shift*.

Shift-share instrument. Formally, let $s_{i,f,t}$ denote pension fund f's share of equity capital in firm i at time t for funds $f = 1, \ldots, F$ and let $M_{f,t}$ denote the total equity holdings of fund f on its balance sheet at time t. Then:

$$Z_{i,2002} = \sum_{f=1}^{F} \underbrace{s_{i,f,2002}}_{\text{share}} \cdot \underbrace{(\ln M_{f,2003} - \ln M_{f,2006})}_{\text{shift}}, \tag{3.3}$$

Here, $Z_{i,2002}$ can be interpreted as the average percentage reduction in pension fund stock holdings of firm i, weighted across funds f by their portfolio shares. The shift $\ln M_{f,2003} - \ln M_{f,2006}$, that is, the percentage divestment from equities, varies across pension schemes because of differences in funding levels prior to the reform.

The key identifying assumption in (3.3) is that fund-level equity sales were uncorrelated with differences in firm-level outcomes post reform. Conversely, identification would be threatened if there was assortative matching between pension funds and firms prior to the reform in ways that affect the differences in post-reform firm performance. An example would be underfunded pension funds selecting on badly performing firms. In Section 3.3.3, I discuss these threats to identification in detail and provide a series of exercises and alternative specifications to rule out potential endogeneity concerns, including an additional instrument for pension schemes' pre-reform funding level.

The specification in equation (3.3), defines the 2001/2002 financial year (1 April 2001 to 31 March 2002) as the base year and measures the shares in (3.3) as of 31 March 2002, denoted by $s_{i,f,2002}$. As discussed, draft legislation for the Pensions Act 2004 was first announced in spring 2003. Using 2002 as the base year is supported by the pension fund flow data. As shown in Figure 3.1 and Appendix D.2, pension funds began selling equities in 2003. Net quantities sold remained small until March 2004. The bulk of equity divestment occurred between autumn 2004 and autumn 2006. To avoid contamination due to the Global Financial Crisis (GFC), I use only the change in equity holdings up to March 2006 in the baseline specification.[13] Appendix D.5 contains a series of robustness checks.

[13] Most papers date the onset of the GFC with the Bear Stearns bailout in June 2007 (e.g. Chodorow-Reich, 2014). Even a more conservative dating, based on New Century's Chapter 11 filing on 2 April 2007, would still justify including the entire financial year 2006/2007 in the analysis.

Reduced-form specification. Using the instrument defined in (3.3), I estimate the effect of the 2002 reform at the firm level on some outcome variable $y_{i,2002+h}$ over some horizon h by running local projections of the form:

$$\ln y_{i,2002+h} - \ln y_{i,2002} = \alpha_h + \beta_h Z_{i,2002} + \boldsymbol{\gamma}_h' \boldsymbol{X}_{i,2002} + \delta_{i,h} \sum_{f=1}^{F} s_{i,f,2002} + \varepsilon_{i,h}, \tag{3.4}$$

where β_h is the coefficient of interest. Note that I have defined $Z_{i,2002}$ in (3.3) over a decrease in holdings. β_h therefore captures the *cumulative* effect of a one percentage point *decrease* in pension fund ownership of firm equity on outcome variable y in year $2002 + h$. Moreover, α_h is a horizon (time) fixed effect and $\boldsymbol{X}_{i,2002}$ is a vector of controls containing book-to-market ratio, market capitalisation, capital structure, maturity structure of outstanding debt, cash holdings, the share of intangible capital, R&D intensity, and firm size, proxied by total employment. The data has been winsorised at the 5th and 95th percentiles. Standard errors are clustered at the firm level. As the shares $s_{i,f,2002}$ do not sum up to one, I control for their sum as recommended by Borusyak et al. (2025). Appendix C.1.2 presents the distribution of shares and Appendix C.1.3 contains the standard balance tests.

Since (3.4) is a shift-share specification, the coefficient of interest β_h should be interpreted as a continuous difference-in-difference between two otherwise identical firms due to a change in pension fund investment exposure, that is, the differential effect of a one percentage point decrease in pension fund investment.

3.3.2 The effect of pension investment at the firm level

I run the regression in (3.4) for a series of outcome variables maintaining the same baseline specification. In the text below, I present the results for firms' stock prices, capital investment, and R&D. In the Appendix I discuss the effect on returns, firms' capital structure, and a measure of investment duration.

Stock prices and returns. To estimate the impact of pension funds' withdrawal from domestic equity markets on valuation, I first run (3.4) using firms' stock prices as the dependent variable. Figure 3.3 shows the cumulative log-change in stock prices relative to the baseline year 2002 before the announcement of the pensions reform. The vertical line indicates the announcement of the reform. There are no significant pre-trends before 2002.

The results suggest that a reduction in pension fund investment is associated with a decrease in stock prices for firms more exposed to pension investment. An additional one percentage point reduction in pension fund equity holdings leads to a 0.5 percent decrease in stock prices one year after the reform. Five years after the shock, stock prices are a further percentage point lower. As I show in the next section, this further drop in stock prices could possibly be driven by changes in firms' fundamentals driven by their reaction to the reduction in valuations.

Pension capital withdrawal has a persistent impact on firm valuations. In a frictionless capital market, one would not expect sales by pension funds to influence prices over such long horizons, since other investors should step in to absorb the drop in demand and stabilise prices. In contrast, my results are consistent with a limits to arbitrage view of equity markets under which pension fund divestment is not fully offset by other investors, for example due to limited entry or inelastic demand, leading to a reduction in market prices for firms more exposed to the shock (Shleifer and Vishny, 1998; Gromb and Vayanos, 2010). An emerging literature documents that demand-driven outflows can have persistent effects on valuations. These effects can be very large, both at the firm (micro) and the aggregate (macro) level (Gabaix and Koijen, 2021).[14] Such persistent demand effects have also been documented in conjunction with other regulatory changes (e.g. Koijen et al., 2024).

In the next section, I show how these price effects altered firms' investment decisions. I first discuss the impact on capital investment and then on R&D.

Capital expenditure. The left panel of Figure 3.4 presents the results of regression (3.4) using capital expenditure over lagged assets as the dependent variable. This regression is meant to capture firms' investment intensity. The results suggest that a reduction in pension fund equity holdings reduces long-term capital investment. Specifically, an additional one percentage point decrease in pension fund equity holdings is, on average, associated with a one percent decrease in capital expenditure after one year. These findings are in line with the empirical literature which has documented that changes in stock prices due to secondary market trading can have significant effects on firms' long-term investment (e.g. Bond et al., 2013, for a survey).

Innovation. To capture firms' innovation intensity, I use their annual research and development (R&D) expenditure over lagged assets as a proxy. In most growth models, higher rates of R&D spending lead to higher innovation rates and are the key indicator of firms' innovation efforts (Aghion et al., 2014). The right panel of Figure 3.4 presents the result

[14]The market microstructure literature typically finds very large multiplier effects of flows on stock valuations (e.g Frazzini et al., 2018). The literature on index inclusion and deletions typically finds smaller price multipliers between 0.5 and 5 (e.g. Chang et al., 2015; Pavlova and Sikorskaya, 2023). That is, a 1 percent reduction in demand lowers valuations by up to 5 percent. Similar-sized effects are also found in the mutual fund-flows literature (Edmans et al., 2012).

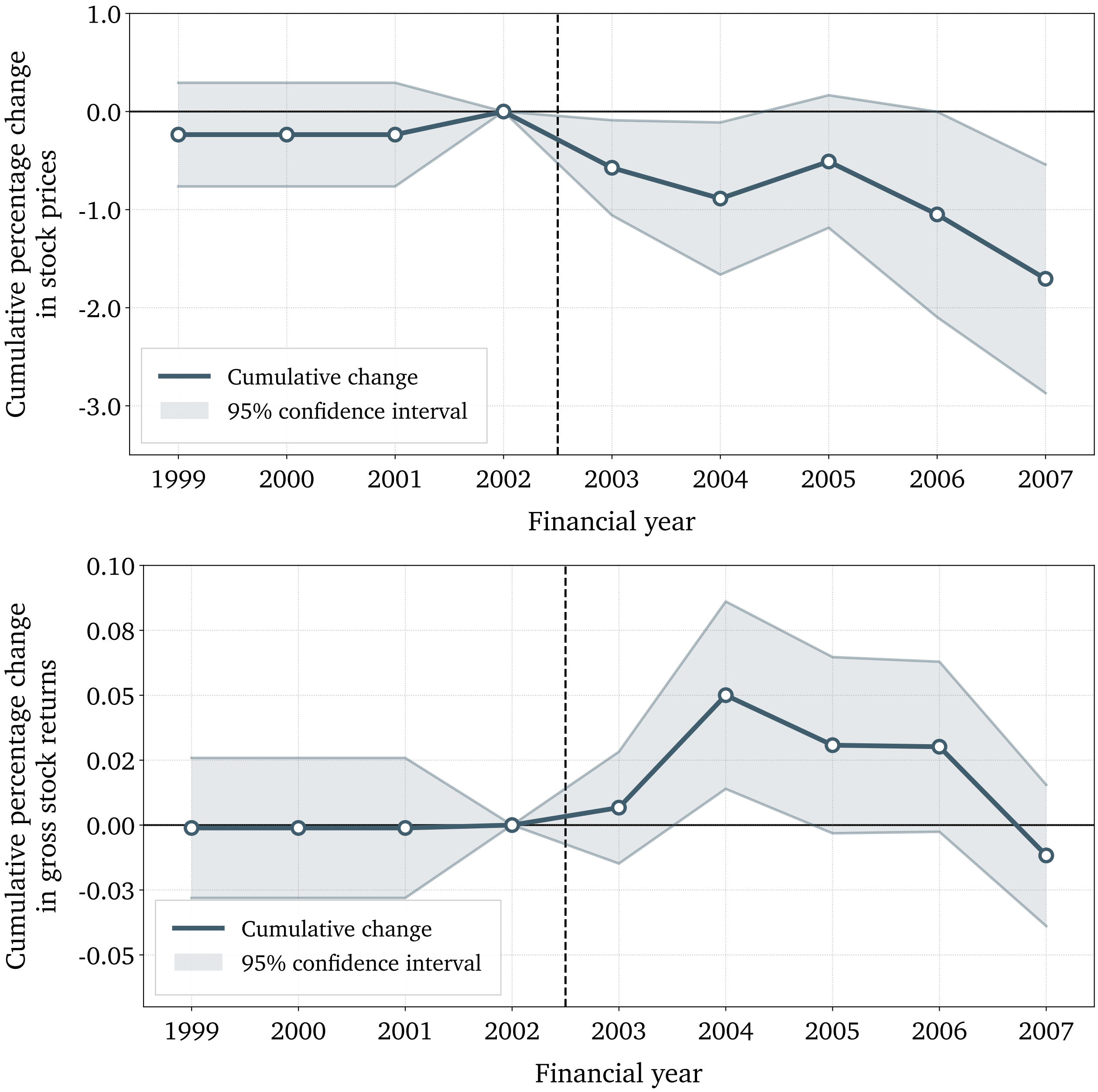

Figure 3.3: Effect on stock prices and realised returns.

The figure shows the cumulative effect of a one percentage point differential reduction in pension fund equity ownership at horizon h as captured by β_h in equation (3.4). The dependent variables are the end-of-year stock price (top panel) and realised gross stock returns (bottom panel).

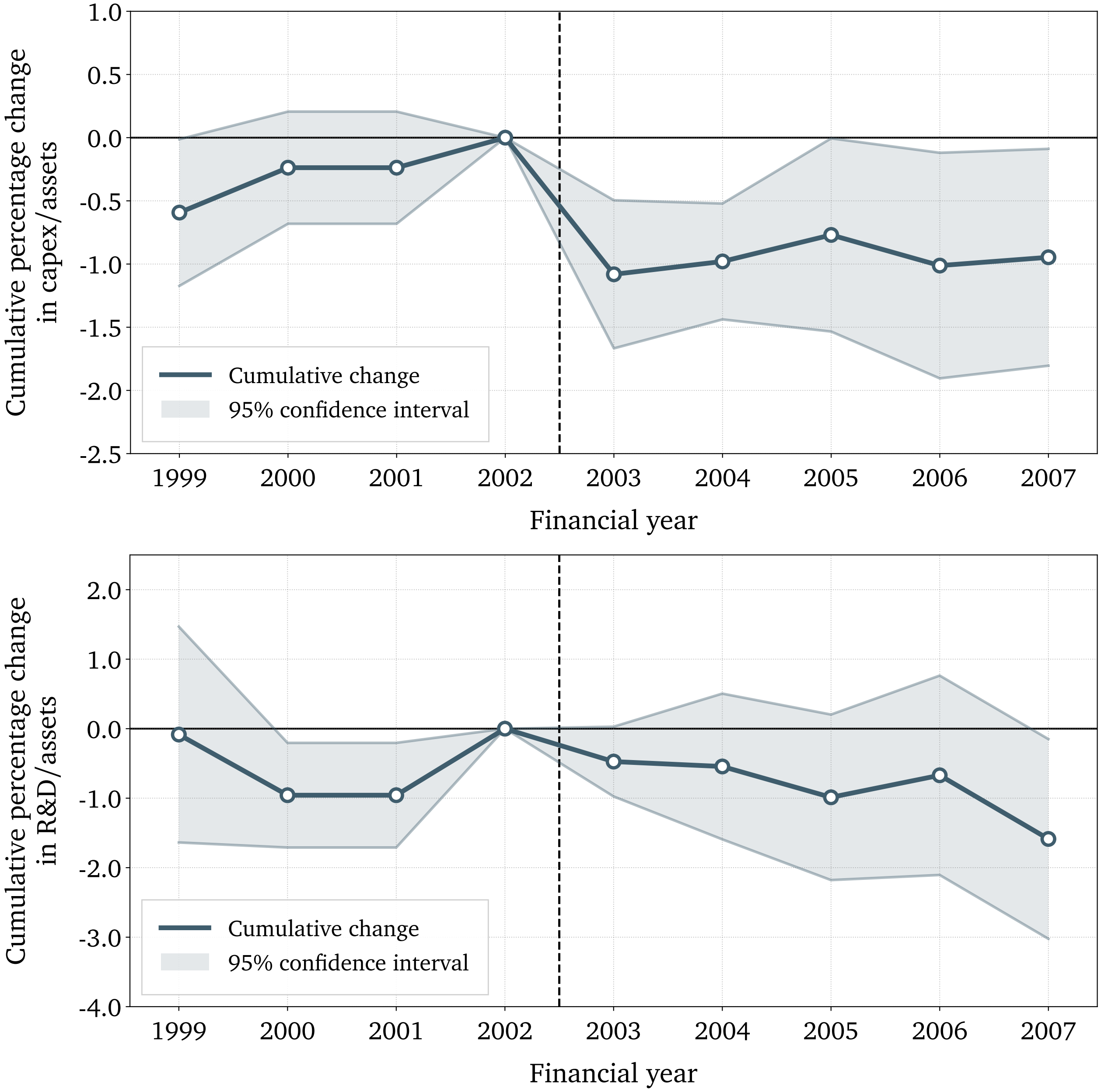

Figure 3.4: Effect on capital and R&D expenditure.

The figure shows the cumulative effect of a one percentage point differential reduction in pension fund equity ownership at horizon h as captured by β_h in equation (3.4). The dependent variables are capital expenditure over assets (top panel) and R&D over assets (bottom panel).

of a regression of the form (3.4) with R&D spending over lagged assets as the dependent variable.

I find evidence that a decrease in pension investment is associated with a lower rate of innovation expenditure. An additional one percentage point reduction in pension fund equity holdings, on average, leads to a 1.2 percent reduction in R&D expenditure over assets at the five-year horizon.

The estimated effects are broadly in line with the literature on the impact of pension investment on innovation spending when accounting for the longer horizon and magnitude of the event study shock. Pozzoli et al. (2024), for example, estimate that having at least one pension fund investor increases firms' patenting rates by about 7 percent and the number of R&D workers by 5 percent. Using an index-inclusion instrument, Bena et al. (2017) find that a one percentage point increase in foreign institutional ownership leads to a 0.3 percent increase in R&D within a year. My results are also in line with other estimates on the impact of changes in stock prices on firms' investment decisions of equity-dependent firms (e.g. Baker et al., 2003; Dong et al., 2012).

3.3.3 Addressing threats to identification

Identification in my empirical strategy comes from the exogenous shifts. As I argued above, pension fund divestment from stocks after the reform is driven by the cross-sectional variation in funding levels between pension schemes prior to the announcement of the reform.

The key identifying assumption is therefore that differences in funding levels between pension schemes are uncorrelated with differences in changes in firm outcomes, such as R&D, capital investment and stock valuations, after the reform. Conversely, identification would be threatened if underfunded pension schemes systematically invested in firms that performed worse post reform, compared to schemes with higher funding levels. The shift-share literature typically performs balance tests on the pre-treatment dependent variable based on the size of the subsequent treatment. These tests can be found in Appendix C.1.3.

In this section, I argue that differences in funding levels are not driven by assortative matching between pension funds and firms. The total amount of equity sales was primarily driven by two factors: on the asset side by the portfolio share of equity investment rather than the composition thereof; and on the liability side by the deterioration in the funding level due to rising future pension commitments.

Equity exposure regression. On the asset side, the consensus view at the time was that funding deficits were largely due to general macroeconomic conditions in the UK (Barclays

Capital Research, 2006), and not specific to individual pension schemes' investment strategies. The downward turn nevertheless differentially affected schemes with a preference to hold large amounts of domestic equity. To test the hypothesis that funding differences were mostly driven by equity exposure rather than scheme quality, I devise the following test:

$$\mathcal{F}_{f,t} = \alpha_t + \beta \text{ ROA}_{f,t} + \delta \text{ UK Equity Share}_{f,t} + \boldsymbol{\gamma}' \boldsymbol{X}_{f,t} + \varepsilon_{f,t}, \tag{3.5}$$

where $\mathcal{F}_{f,t}$ is the funding level at the end of a year, $\text{ROA}_{f,t}$ is the return on assets during that year, and $\boldsymbol{X}_{f,t}$ are fund controls listed in the Appendix. If it is indeed the case that the differences in funding levels are driven by equity exposure, past returns should not be predictive after controlling for the schemes' asset composition. The results reported in Table D.2 in Appendix D.3.2 confirm this hypothesis suggesting that funding differences come primarily from the loading on equities.

Portfolio concentration. The result that the equity share is sufficient to predict the funding level is not surprising. In general, the variation in stocks held between pension schemes is low. Equity investment is concentrated around the benchmark FTSE-100 or FTSE-350. Even though some pension funds deviate from the benchmark, there is no systematic overweighting of certain stocks by funds with low funding levels prior to the reform.

Table C.2 in Appendix C.3 reports various measures of portfolio concentration. The standard deviation of the Herfindahl-Hirschman index of portfolio concentration, for instance, is only 0.01, indicating little variation in portfolio weights. The cosine similarity, a common measure of similarity of two vectors is on average 0.8, again indicating very high similarity between portfolios. This confirms that most variation across funds comes from equity portfolio shares rather than within-equity variation of portfolio allocations.

Membership structure. On the liability side, the key determinant of funding status is a scheme's future pension commitment and hence its membership structure. A pension scheme with few active contributors and many pensioners faces higher liabilities while collecting less in contributions. Hence, its funding level should be systematically lower. I leverage this insight in a final robustness check. As I show in Appendix D.5.4 and Figure D.5.4, funding levels across pension schemes are highly correlated with the ratio of contributors relative to pensioners. In fact, the reduction in equity investment post reform can be predicted by a scheme's pre-reform membership structure. In conjunction with the results in the previous section, this suggests that pre-reform funding levels were driven in large part by the liability side of schemes' balance sheet and their U.K. equity exposure.

The results from the membership instrument regression reported in Appendix D.5.4 confirm the baseline results: Firms more exposed to pension fund investment experience a drop in stock prices, and subsequently invest less in R&D.

4 Quantifying the impact on economic growth

In this section, I quantify the growth impact of pension investment using the empirical estimates from the natural experiment and the theoretical model from the previous section. I then simulate the reform via a tightening of risk-constrained investors' risk-bearing constraint. Finally, I decompose the macro effect into its various channels.

4.1 Calibration

I calibrate the model at annual frequency. The model features 11 structural parameters. I identify these parameters using a mixture of calibration and indirect inference with simulated method of moments (SMM). Table 4.1 contains an overview. For the calibration, I split the data in three periods, a pre-reform period from 1985 to 2002, a reform period from 2002 to 2007, and a post-reform period from 2007 onwards. For time series moments, the baseline for my calibration is the pre-reform period, while for cross-sectional moments it is the financial year 2001-2002, the last before the announcement.[15]

The quantitive simulation also requires taking a stance on the entry cost in (2.17). For the baseline I choose a power function:

$$\psi(e) = e^{\psi}, \tag{4.1}$$

where the parameter ψ will be calibrated internally as explained below.

Normalisations. I normalise household labour supply to $L = 1$. All aggregate variables in the model can therefore be interpreted as per-capita values. The parameter κ captures the tightness of the regulatory constraint on risk-constrained investors. As the parameter is only identified relative to γ, I normalise its initial level at $\kappa = 1$. A calibration with $\kappa = 0$ would be isomorphic as only changes in risk aversion will affect the quantitative results. Any changes in levels can be accommodated by different levels of risk aversion γ^M and γ^C.

[15]For the entirety of the quantitative section, I have removed the financial crisis when calculating time-series moments.

Table 4.1: Calibrated parameters.

#	Parameter	Description	Identification	Value
1	L	Labour	Normalisation	1.00
2	κ	Value-at-risk constraint	Normalisation	1.00
3	η	R&D curvature	External	1.75
4	β	Risk-constrained share	External	0.25
5	γ	Investor risk aversion	External	1.33
6	r	Discount rate	External	0.05
7	ϑ	Elasticity of substitution	External	0.60
8	σ_a	Volatility shifter	Internal	0.55
9	ν	Step size outsiders	Internal	1.61
10	ψ	Entry cost	Internal	3.56
11	ζ	R&D cost shifter	Internal	1.25

External calibration. First, the parameters η and ϑ are standard features of innovation models. ϑ is the Cobb-Douglas weight on labour in the production function and hence the labour share of income. Typical estimates put $\vartheta = 0.6$. The parameter η governs the curvature of the R&D cost function for incumbent firms. The growth literature typically uses a value between 2 and 3. Akcigit and Kerr (2018) use 2.5, but there is evidence that a value closer to 1.5 is appropriate for larger firms (Dechezleprêtre et al., 2023). As a baseline quantification, I set $\eta = 1.75$.

Second, the parameters r and β are readily observable in the data. Because of linear utility (2.1), the household's discount rate r is the equilibrium interest rate in the model. The average real interest rate in the U.K. for the period from 1985 to 2002 is around 5 percent. Following the empirical results in the paper, I interpret *risk-constrained investors* to refer to *defined-benefit pension funds*.

The Office of National Statistics (ONS) collects data on the aggregate shareholding in listed British companies by shareholder type. For the pre-reform period, the average fraction of U.K. shares held by DB pension schemes is 12 percent. In the model, this fraction is symmetric across stocks and given by $\beta X_t^C / [(1-\beta) X_t^C + \beta X_t^M] = \beta \alpha^C W_t^C / Q_t$, see (2.20). Given an equity portfolio share of 47 percent in the initial steady state, this implies $\beta = 0.12/0.47 \approx 0.25$. In equilibrium, the portfolio share is given by (2.13), but using the risk

Table 4.2: Targeted moments.

#	Moment	Model	Data	Method	Weight
1	Real interest rate	0.05	0.05	External	n/a
2	Labour income share	0.60	0.60	External	n/a
3	Share of DB pension fund stock holdings	0.12	0.12	External	n/a
4	U.K. equity portfolio share	0.47	0.47	External	n/a
5	Growth rate	0.028	0.028	Internal	1.00
6	Entry rate	0.041	0.035	Internal	1.00
7	R&D elasticity	1.200	1.200	Internal	1.00
8	Equity risk premium	0.067	0.045	Internal	1.00

premium (2.23) the portfolio share collapses to the fully parametric expression. I back out $\gamma = 1.33$ which supports the observed asset allocation.

Internal calibration. The remaining four parameters $\Gamma = \{\underline{\sigma}, \nu, \psi, \zeta\}$ are calibrated to jointly reproduce five moments based on U.K. macro data and my empirical results in Section 3.3. I use a Simulated Method of Moments (SMM) approach. Specifically, the procedure involves searching over the parameter space $\Gamma \in \boldsymbol{h}(\Gamma)$ to minimise the distance between model generated moments $\boldsymbol{\mathcal{M}}(\Gamma)$ and their empirical counterparts $\boldsymbol{\mathcal{M}}_0$. The objective function is:

$$\min_{\Gamma \in \boldsymbol{h}(\Gamma)} ||\boldsymbol{W}\left(\boldsymbol{\mathcal{M}}(\Gamma) - \boldsymbol{\mathcal{M}}_0\right)||_p, \tag{4.2}$$

where $\boldsymbol{W}$ is a weighting matrix, $||\cdot||_p$ denotes the p^{th} norm, and $\boldsymbol{h}(\Gamma)$ is the admissible parameter space. In the baseline calibration, I set weights $[1, 1, 1, 1]$ across moments and use the Euclidean norm ($p = 2$).[16]

The targets are chosen as follows: First, I target the average annual growth rate of the economy in the pre-pension reform period. I choose January 1985 as the start and December 2002 as the end date. Using a conventional Hodrick-Prescott filter, the average annualised growth rate is 2.80 percent.

Second, in general equilibrium, the effects of any firm-level policy propagates via the

[16]The optimisation routine is based on a generalised pattern search algorithm, which is generally more robust than gradient-based methods when calibrating endogenous growth models with a high degree of linearity.

labour market (2.19) to output and firm entry. To scale the importance of entry innovation for growth, I target an entry rate of 4.5 percent per year. This implicitly also pins down the size of the final good sector.

The results in Section 3.3.2 provide an estimate for the firm-level response of R&D investment to a change in the share of firm equity held by defined-benefit pension funds. The point estimate for the steady-state-to-steady-state *semi-elasticity of R&D spending* over total assets with respect to a one percentage point decrease in equity holdings is 1.2. The model counterpart is $\zeta z_{it}^{\eta}(q_{it}/Q_t)w_t$. As there is no real notion of assets in the model, I divide this expression by q_{it} to proxy for firm size. I capture the effect of the pension reform via a change in the parameter κ, the tightness of the regulatory constraint. In the model, the semi-elasticity is:

$$\varepsilon \equiv -\frac{1}{N}\sum_{i=1}^{I}\frac{d}{d\kappa}\ln\left(\frac{\zeta z_{it}^{\eta}w_t}{Q_t}\right). \tag{4.3}$$

Note that this approach implies that only the firm-level response to a change in the parameter κ is targeted, but that the aggregate elasticity of the growth rate remains untargeted.

Finally, I match the average equity risk premium on British equities for the period between 1985 and 2001, which is 4.41 percent (Damodaran, 2019, p.38). Table 4.2 contains a list of targeted moments in the model and the data. Further notes on the calibration are in Appendix E.1, including untargeted moments and sensitivity analysis.

4.2 Calibrated equilibrium

Untargeted moments. The model-implied average annual return on firms' stock μ is 15.8 percent. The risk premium is approximately 6.7 percent, compared to 4.41 in the data. The implied market volatility is 20.7 percent per year, or 5.96 percent per month. Damodaran (2019, p.38) reports an annual volatility of 20.03 percent (5.78 percent per month) for the period between 1981-2001.

Labour used in the production of the final good is $L^P = 0.85$, which implies that around 15 percent of workers are employed in the research sector. Total research spending is defined as $\sum_{i=1}^{N}\left[\psi e_{it} + \zeta z_{it}^{\eta}\right]\varphi_{it}w_t/Y_t$ and accounts for 20 percent of GDP, which is much larger than in the data but not surprising given that the model is a Schumpeterian growth model that abstracts from capital accumulation. Finally, at the firm level, the model implies a price earnings ratio of $P_t q_{it}/(\pi_t q_{it} - \zeta z^{\eta} w_t \varphi_{it})$ of approximately 10.4, which is at the lower end of estimates for the U.K. in the late 1990s and early 2000s.

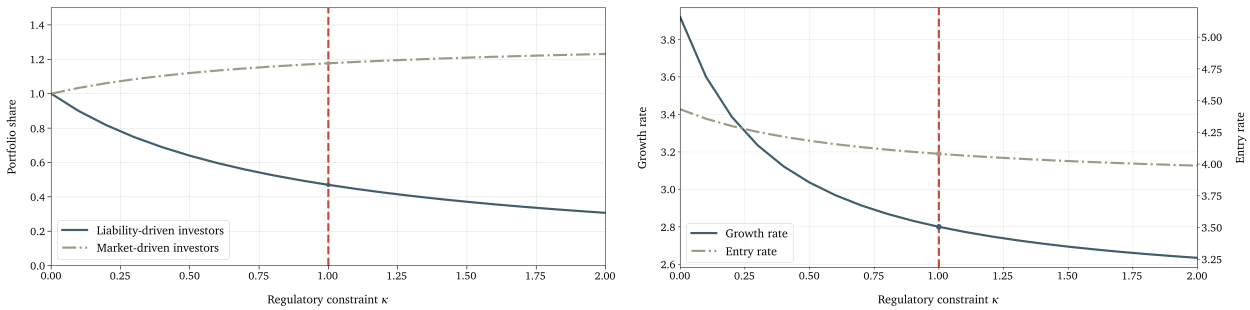

Figure 4.1: Equilibrium portfolio shares, innovation, and growth.

The left panel shows the equilibrium portfolio shares for risk-constrained and market-driven investors as a function of the regulatory constraint parameter κ. The red line denotes the calibrated pre-reform steady state. The right panel shows entry innovation (dashed line, right axis) and growth (solid line, left axis) as function of the regulatory constraint parameter κ.

The aggregate effects of regulation. The key parameter of interest is the tightness of the regulatory constraint κ on the volatility of risk-constrained investors' portfolios, which affects growth through investors' risk premium. To illustrate how the key endogenous variables in the model behave as a function of κ, I solve the model for a range of values.

Figure 4.1 shows that a tighter regulatory constraint on risk-constrained investors lowers their equity portfolio share. In the pre-reform steady state, approximately 47 percent of their wealth is allocated to equity. To ensure market clearing, market-driven investors short the safe bond. Their portfolio share is larger than one.

As the regulatory constraint tightens, risk-constrained investors reduce their stock holdings in incumbent firms. The market risk premium rises to incentivise market-driven investors to buy and clear the market. Consequently, incumbent innovation falls. Although lower investment rates imply lower volatility in firms' quality level, this feedback channel between investment and the risk premium is not sufficiently strong to compensate for the fall in investor demand due to the tighter regulatory constraint.

While the sign on incumbent innovation is clearly determined in theory, the entry effect is ambiguous. Although lower incumbent innovation frees up resources that can be used for entry, lowering the wage and therefore the cost of market entry, a higher risk premium can also reduce the value of becoming an incumbent firm in the first place. This second indirect effect dominates the (positive) reallocation effect in the quantitative model. The net effect on entry is negative. The model therefore predicts that a reduction in the demand for incumbent firms' equity has a net negative effect on firm creation. The joint response determines the effect on economic growth, as illustrated in the right panel of Figure 4.1.

Table 4.3: Comparative statics.

Parameter value	**Growth effect**	**Entry response**
$\eta = 1.35$	0.06	negative
$\eta = 1.75$	0.14	negative
$\eta = 2.00$	0.19	positive

4.3 The growth effects of risk-limits

Approach. In the final part of the paper, I evaluate the conditions under which the impact on growth is large. The spirit of this exercise is similar to Aghion et al. (2025). Specifically, holding constant all other model parameters, I vary the tightness of the regulatory constraint κ to match the observed decline in the equity portfolio share for DB pension funds from 2002 to 2007. As the model does not feature transition dynamics, I interpret every year as a new steady state. I then evaluate under which conditions the model produce a large decline in growth and decompose the relative contributions of incumbent and entry innovation.

Simulation. Given a sequence of pension fund equity portfolio shares observed in the data, $\boldsymbol{\alpha}$, I invert the first-order condition (2.13) to back out a vector of implied regulatory tightness $\boldsymbol{\kappa}$ that rationalises these portfolio shares. Given that the model features aggregate risk, I run a *Monte-Carlo simulation* of 100 firms with 10,000 draws for each steady state. In the baseline simulation, the model predicts a decline in the annual expected growth rate from 2.80 percent, the pre-reform mean for the period from 1985 to 2002, to 2.66 percent, that is, a reduction of 0.14 percentage points.

Table 4.3 illustrates that the size of the growth effect crucially depends on the elasticity of firm investment with respect to changes in the value of innovation η. I recalibrate the model, maintaining all baseline calibration targets, and repeat the exercise from above. When the elasticity η is low, incumbent firms do not react to changes in stock prices and the growth effects are small. However, the effect on entry becomes more pronounced and negative. The intuition is that for low incumbent investment elasticities, the drop in labour demand is small and therefore the reallocation effect is weak, while the risk premium effect (the change in stock prices) directly affects entry innovation. In the scenario with high investment elasticity, the reallocation effect is sufficiently strong such that it compensates for the risk premium effect. The aggregate impact on the growth rate, however, remains negative as incumbent innovation drops even further.

5 Conclusion

This paper asks how financial regulation that limits investors' capacity to take risk affects economic growth. I develop a theory of segmented equity markets and endogenous growth. Firms maximise their stock market valuation and therefore consider how risky investment choices feed back into valuations via risk premia. When the marginal investor becomes more risk averse, risk premia have to rise to clear the market. Firms react by cutting back on high-risk investment.

These results suggest that shifts in investor demand can have significant effects on firms' investment choices. Motivated by these findings, I study a particular example of plausibly regulation-driven changes in investor demand in British equity markets. I show that a reduction in equity demand reduced valuations and investment for firms more affected by the reform. I interpret these results as evidence that market segmentation and hence the ability of equity investors to absorb risk are important determinants of real outcomes.

A calibrated version of the model suggests that the growth impact can sizeable when firms are sensitive to changes in valuations and when investor demand curves are inelastic.

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Supplementary Appendix

The Impact of Pension Investment on Economic Growth

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Contents

A Appendix – Proofs and Derivations

A.1 Baseline model

A.1.1 Household

The household holds bonds A_t, supplies L units of labour at wage w_t, earns V_t from investors and consumes dC_t. The household maximises (2.1) subject to the budget constraint $dA_t = rA_t dt + w_t L dt + V_t dt - dC_t$. Because utility is linear, the bond market clears at the discount rate r.

A.1.2 Production

Final good. The competitive final good producer maximises (2.2) subject to $Y = w_t L_t^P + \sum_{i=1}^N \varrho(q_{it}) y_{it}$ by choosing labour L_t^P and intermediate inputs $\{y_{it}\}_{i=1}^N$. w_t is the wage and $\{\varrho(q_{it})\}_{i=1}^N$ are the prices of the available inputs of quality $\{q_{it}\}_{i=1}^N$. The solution is (2.3).

Intermediate goods. Intermediate firms take the demand function (2.3) for their inputs as given. Marginal cost are $1-\vartheta$ units of the final good. Firms choose prices $\varrho(q_{it})$ to solve:

$$\max\Big\{\varrho(q_{it})y_{it} - (1-\vartheta)y_{it}\Big\} \qquad \text{s.t.} \qquad y_{it} = \varrho(q_{it})^{-\frac{1}{\vartheta}} q_{it} L_t^P. \tag{A.1}$$

The first-order condition with respect to the price for a given quality can be written as $(1-1/\vartheta)\,\varrho(q_{it})^{-1/\vartheta} q_{it} L_t^P - (1-\vartheta)/\vartheta\,\varrho(q_{it})^{-1/\vartheta-1} q_{it} L^P = 0$ or equivalently $\varrho(q_{it}) = 1$. Substituting back into (A.1) yields

$$\varrho(q_{it}) = 1, \qquad y_{it} = L_t^P q_{it}, \qquad \text{and} \qquad \pi(q_{it}) = \vartheta L_t^P q_{it}. \tag{A.2}$$

Aggregation. Define aggregate quality Q_t as in (2.4). Substitute $y_{it} = L_t^P q_{it}$ back into (2.2) to get $Y_t = L_t^P Q_t/(1-\vartheta)$. Adding up over firms yields profits $\Pi_t = \vartheta L_t^P \sum_{i=1}^N q_{it} = \vartheta L_t^P Q_t$ and the final good used in production $Y_t^P = (1-\vartheta)\sum_{i=1}^N y_{it} = (1-\vartheta) L_t^P Q_t$. Finally, given labour demand (2.3), the wage is $w_t = \vartheta Q_t/(1-\vartheta)$. We can check the identity $Y_t = Y_t^P + \Pi_t + w_t L_t^P = (1-\vartheta)L_t^P Q_t + \vartheta L_t^P Q_t + \frac{\vartheta}{1-\vartheta} L_t^P Q_t = \frac{1}{1-\vartheta} L_t^P Q_t$. Net output is $Y_t - Y_t^P = \vartheta(2-\vartheta)/(1-\vartheta) L_t^P Q_t$.

A.1.3 Portfolio choice

Investors solve their optimisation problems (2.9) and (2.12). Note that the law of motion for changes in instantaneous wealth 2.8 has two sources of shocks, aggregate risk $d\mathcal{Z}_t \sim \mathcal{N}(0, dt)$ and creative destruction jumps $d\mathcal{J}_{it} \sim Poisson(e_{it})$. The instantaneous variance of investors' wealth is $\mathrm{Var}\left[dW_t^M/W_t^M\right] = \sum_{i=1}^N \left(\alpha_{it}^M\right)^2 e_{it}dt + \mathrm{Var}\left[\sum_{i=1}^N \alpha_{it}^M \varsigma_{a,it} dZ_t^a\right]$. Noting that aggregate shocks are perfectly correlated across investors, one can re-write the expression as $\mathrm{Var}\left[dW_t^M/W_t^M\right] = \sum_{i=1}^N \left(\alpha_{it}^M\right)^2 e_{it}dt + \sum_{i=1}^N \sum_{j=1}^N \mathrm{Cov}\left(\alpha_{it}^M \varsigma_{a,it} d\mathcal{Z}_t, \alpha_{jt}^M \varsigma_{a,jt} d\mathcal{Z}_t\right) = \sum_{i=1}^N \left(\alpha_{it}^M\right)^2 e_{it}dt + \left[\sum_{i=1}^N \left(\alpha_{it}^M \varsigma_{a,it}\right)^2 + 2\sum_{i<j} \alpha_{it}^M \alpha_{jt}^M \varsigma_{a,it}\varsigma_{a,jt}\right] dt$ or equivalently we can write $\mathrm{Var}\left[dW_t^M/W_t^M\right] = \sum_{i=1}^N \left(\alpha_{it}^M\right)^2 e_{it}dt + \left(\sum_{i=1}^N \alpha_{it}^M \varsigma_{a,it}\right)^2 dt$, where the subscript $i < j$ in the sum operator denotes the sum over all distinct unordered pairs to avoid double counting. The investor's problem can be written as:

$$\begin{aligned}\max_{\{\alpha_{it}^M\}} rW_t^M dt &+ \sum_{i=1}^N \alpha_{it}^M \left(\mu_{it} - r - e_{it}\right) W_t^M dt \\ &- \frac{\tilde{\gamma}}{2}\left[\sum_{i=1}^N \left(\alpha_{it}^M\right)^2 e_{it}dt + \left(\sum_{i=1}^N \alpha_{it}^M \varsigma_{a,it}\right)^2 dt\right] \left(W_t^M\right)^2.\end{aligned} \tag{A.3}$$

The first-order condition associated with the problem above is given by $\left(\mu_{it} - r - e_{it}\right) W_t^M dt - \frac{\tilde{\gamma}}{2}\left[2\alpha_{it}^M e_{it}dt + 2\left(\sum_{i=1}^N \alpha_{it}^M \varsigma_{a,it}\right) dt\right] \left(W_t^M\right)^2 = 0$. Risk-constrained investors' problem translates *mutatis mutandis*:

$$\begin{aligned}\max_{\{\alpha_{it}^C\}} rW_t^C dt &+ \sum_{i=1}^N \alpha_{it}^C \left(\mu_{it} - r - e_{it}\right) W_t^C dt \\ &- \left(\frac{\tilde{\gamma}}{2} + \tilde{\kappa}\right)\left[\sum_{i=1}^N \left(\alpha_{it}^C\right)^2 e_{it}dt + \left(\sum_{i=1}^N \alpha_{it}^C \varsigma_{a,it}\right)^2 dt\right] \left(W_t^C\right)^2.\end{aligned} \tag{A.4}$$

The first-order condition associated with the problem above is given by $\left(\mu_{it} - r - e_{it}\right) W_t^C dt - \left(\frac{\tilde{\gamma}}{2} + \tilde{\kappa}\right)\left[2\alpha_{it}^C e_{it}dt + 2\left(\sum_{i=1}^N \alpha_{it}^C \varsigma_{a,it}\right) dt\right] \left(W_t^C\right)^2 = 0$. Investor demand functions are given by the following two expressions:

$$\begin{aligned} X_{it}^M &= \frac{\mu_{it} - r - e_{it} - \gamma\varsigma_{a,it}\left(\sum_{j\neq i}^N X_{jt}^M \varsigma_{a,jt}\right)}{\gamma\left(e_{it} + \varsigma_{a,it}^2\right)}, \\ X_{it}^C &= \frac{\mu_{it} - r - e_{it} - (2\kappa + \gamma)\varsigma_{a,it}\left(\sum_{j\neq i}^N X_{jt}^C \varsigma_{a,jt}\right)}{(2\kappa + \gamma)\left(e_{it} + \varsigma_{a,it}^2\right)}. \end{aligned} \tag{A.5}$$

A.1.4 Stock price and investment

Firms maximise their stock price. In equilibrium, investor risk aversion gives rise to a risk premium. In the model with aggregate risk, this risk premium is affected by investor tilting across stocks in response of firms' loading on aggregate shocks, which in turn depends on their stock prices via (2.7). Hence, firms stock price equations depend implicitly on investors' entire portfolio allocation across all N stocks. Imposing market clearing in each stock market, $\beta X_{it}^C + (1-\beta)X_{it}^M = 1$, yields

$$\frac{\frac{\mu_{it}-r-e_{it}}{\lambda} - \varsigma_{a,it}\left[\beta \sum_{j\neq i}^{N} X_{jt}^C \varsigma_{a,jt} + (1-\beta)\sum_{j\neq i}^{N} X_{jt}^M \varsigma_{a,jt}\right]}{e_{it} + \varsigma_{a,it}^2} = 1, \tag{A.6}$$

where is weighted market risk aversion $\lambda \equiv \gamma(2\kappa+\gamma)/[2\kappa(1-\beta)+\gamma]$ as defined in (2.15). Defining the term in brackets as $\mathcal{B}_{it} \equiv \beta \sum_{j\neq i}^{N} X_{jt}^C \varsigma_{a,jt} + (1-\beta)\sum_{j\neq i}^{N} X_{jt}^M \varsigma_{a,jt}$, one can write (A.6) as:

$$\mu_{it} - r = (1+\lambda)e_{it} + \lambda\varsigma_{a,it}^2 + \lambda\varsigma_{a,it}\mathcal{B}_{it}. \tag{A.7}$$

Substituting the expressions in (2.7) for μ_{it} and $\varsigma_{a,it}$, one can recover the firm's Hamilton-Jacobi-Bellman equation from market clearing in the stock market:

$$\max_{z_{it}} \left\{ \begin{array}{c} \pi(q_{it}) - \zeta z_{it}^{\eta}\varphi_{it}w_t + P'(q_{it})z_{it}q_{it} + \frac{1}{2}P''(q_{it})\sigma_a^2 q_{it}^2 \\ -\lambda\varsigma_{a,it}\mathcal{B}_{it}P(q_{it}) - \left[r + \lambda\left(\varsigma_{a,it}\right)^2 + (1+\lambda)e_{it}\right]P(q_{it}) \end{array} \right\} = 0. \tag{A.8}$$

Note that the aggregate factor $\mathcal{B}_{it}$ drags down the firm's valuation proportional to market risk aversion. Using the definition of the coefficient $\varsigma_{a,it}$ in (2.7), one can rewrite $\lambda\varsigma_{a,it}\mathcal{B}_{it}P(q_{it}) = \lambda\sigma_a\mathcal{B}_{it}P'(q_{it})q_{it}$. It then becomes clear that the exposure to aggregate risk effectively introduces a downward drift in the firm's valuation, the extend of which depends on how much firms quality growth and hence its size adds to the weighted average exposure of investors to the aggregate risk factor, captured by the term $\mathcal{B}_{it}$.

The firm maximises (A.8). As the additional terms related aggregate risk cannot be affected by the firm's choice of drift z_{it}, these terms only enter the first-order condition via the stock value $P(q_{it})$.

A.1.5 Market clearing conditions

The markets in the economy in the economy are the final good, labour, and bonds, plus N markets for intermediate goods and stocks each. For the first three, we have:

$$Y_t = Y_t^P + dC_t, \tag{A.9}$$

$$L = L_t^P + \sum_{i=1}^{N} \psi\left(e_{it}\right)\varphi_{it} + \sum_{i=1}^{N} \zeta z_{it}^{\eta}\varphi_{it}, \tag{A.10}$$

$$A_t = (1-\beta)\left(1 - \sum_{i=1}^{N} \alpha_{it}^{M}\right) W_t^M + \beta\left(1 - \sum_{i=1}^{N} \alpha_{it}^{C}\right) W_t^C, \tag{A.11}$$

and for $i = 1, \ldots, N$ intermediate goods and stocks:

$$y_{it} = \varrho(q_{it})^{-\frac{1}{\vartheta}} L_t^P q_{it}, \tag{A.12}$$

$$Q_t = (1-\beta)X_{it}^M + \beta X_{it}^C. \tag{A.13}$$

A.1.6 Firm size distribution

Let $p_t(q)$ be the distribution of product qualities. In a symmetric equilibrium with $e_{it} = e_t$, its evolution is governed by a standard Kolmogorov-forward equation:

$$\frac{\partial p_t(q)}{\partial t} = -\frac{\partial}{\partial q}\left[z_t q_t p_t(q)\right] + \frac{1}{2}\cdot\frac{\partial^2}{\partial q^2}\sigma_a^2 q^2 p_t(q) + e_t\left[\frac{1}{\nu}p_t\left(\frac{q}{\nu}\right) - p_t\left(q\right)\right]. \tag{A.14}$$

A.2 Equilibrium

As explained in the main text, I drop indexes to indicate symmetry across firms and guess that a firm's price function is linear in the quality level $P(q) = Pq$, which implies that stock price volatility is the same as the underlying volatility of the quality level $\varsigma_a = \sigma_a$. Moreover, on a balanced growth path all aggregate variables grow at a common constant growth rate, the growth rate of aggregate quality Q. Hence, I define the wage-to-quality ratio $\omega \equiv w/Q$.

With investor demand in (A.5) a symmetric equilibrium requires demand function to take the form:

$$X^M = \frac{\mu - r - e}{\gamma\left(N\sigma_a^2 + e\right)} \qquad \text{and} \qquad X^C = \frac{\mu - r - e}{(2\kappa+\gamma)\left(N\sigma_a^2 + e\right)}. \tag{A.15}$$

Denoting the wage relative to aggregate quality by ω as before, the firm's price function

(A.8) collapses to:

$$\left[r + (1+\lambda)e + \lambda\sigma_a^2 N\right] P = \max_z \ \pi - \zeta z^\eta \omega + zP, \tag{A.16}$$

where the new term $\lambda\sigma_a^2 N$ parametrises the aggregate component on the risk premium. Moreover, I use the short-hand notation $\pi = \vartheta L^P$ to denote the firm's profit flow. The equilibrium risk premium is:

$$\mu - r = (1+\lambda)e + \lambda N\sigma_a^2. \tag{A.17}$$

As labour demand in Appendix A.1.2 implies $\omega = \vartheta/(1-\vartheta)$, we obtain the expressions in the main text. General equilibrium is then described by the price equation (A.16), the first-order condition 2.16, the free-entry condition (2.17) and the labour market clearing condition (2.19).

A.3 Comparative statics

The model is described by five equations in five unknowns, the entry rate e, the rate of incumbent innovation z, the risk premium μ, labour used in prodcution of the final good L^P, and the stock price P. The full system is:

$$z = \left[\frac{(1-\vartheta)P}{\zeta\eta\vartheta}\right]^{\frac{1}{\eta-1}} \tag{A.18.i}$$

$$P = \frac{\vartheta}{\mu+z}\left(L^P - \frac{\zeta}{1-\vartheta}z^\eta\right) \tag{A.18.ii}$$

$$\mu = r + (1+\lambda)e + \lambda N\sigma_a^2 \tag{A.18.iii}$$

$$L^P = L - \psi(e) - \zeta z^\eta \tag{A.18.iv}$$

$$P = \frac{\vartheta}{(1-\vartheta)\nu} \cdot \frac{\psi(e)}{e} \tag{A.18.v}$$

Note that any change in the regulatory constraint κ maps directly into a change in market risk aversion λ. Substituting (A.18.iii) and (A.18.iv) into (A.18.ii), then differentiating the system with respect to z, P and e as well as the parameter λ yields:

$$\hat{z} - \frac{z}{(\eta-1)P}\hat{P} = 0, \tag{A.19.i}$$

$$\left(\frac{\zeta\eta\vartheta z^{\eta-1}}{1-\vartheta} - P + \vartheta\zeta\eta z^{\eta-1}\right)\hat{z} + \left[r + (1+\lambda)e + \lambda N\sigma_a^2 - z\right]\hat{P}$$
$$+\left[(1+\lambda)P + \vartheta\psi'(e)\right]\hat{z}^e + \left(e + N\sigma_a^2\right) = 0, \tag{A.19.ii}$$

$$\nu e\hat{P} + \left[\nu P - \frac{\vartheta}{1-\vartheta}\psi'(e)\right]\hat{e} = 0, \tag{A.19.ii}$$

where $\hat{z} \equiv dz/d\lambda$, $\hat{P} \equiv dP/d\lambda$ and $\hat{e} \equiv de/d\lambda$. Note that $\frac{\zeta\eta\vartheta z^{\eta-1}}{1-\vartheta} - P = 0$ via the envelope condition in (A.18.i). Use (A.18.iii) and write the system as:

$$\begin{bmatrix} \hat{e} \\ \hat{P} \\ \hat{z} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{\left(e+N\sigma_a^2\right)P}{\mu-z} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{\nu e}{\nu P - \frac{\vartheta}{1-\vartheta}\psi'(e)} & 0 \\ -\frac{(1+\lambda)P+\vartheta\psi'(e)}{\mu-z} & 0 & -\frac{(1-\vartheta)P}{\mu-z} \\ 0 & \frac{z}{(\eta-1)P} & 0 \end{bmatrix} \begin{bmatrix} \hat{e} \\ \hat{P} \\ \hat{z} \end{bmatrix}. \tag{A.20}$$

Write the system as $x = b + Ax$. Cramer's rule allows to directly solve for each element i using $x_i = det(A_i)/det(A)$ where A_i is the matrix formed by replacing the i^{th} column of A by the vector b. The determinant of the matrix system is given by $det(A) = \nu e\left[(1+\lambda)P + \vartheta\psi'(e)\right] - \left(\nu P - \frac{\vartheta}{1-\vartheta}\psi'(e)\right)\left(\mu - z + \frac{(1-\vartheta)z}{\eta-1}\right)$. Define $\Lambda \equiv -det(A)$. We have:

$$\frac{\hat{P}}{P} = -\frac{\nu P - \frac{\vartheta}{1-\vartheta}\psi'(e)}{\Lambda/(e+N\sigma^2)}, \qquad \frac{\hat{e}}{e} = \frac{\nu P}{\Lambda/\left(e+N\sigma^2\right)} \qquad \text{and} \qquad \frac{\hat{z}}{z} = -\frac{\hat{P}}{(\eta-1)P}. \tag{A.21}$$

Note that $\Lambda > 0$ as a negative value would imply a negative stock price P and therefore no equilibrium to the model. The response of incumbent innovation z inherits its sign directly from the stock price. There are two relevant cases.

Negative entry response. When $\psi'(e)\frac{\vartheta}{1-\vartheta} > \nu P$, it follows directly that $\Lambda < 0$ and $\hat{P} < 0$, $\hat{z} < 0$, as well as $\hat{e} < 0$. On net, entry becomes less attractive.

Positive entry response. When $\psi'(e)\frac{\vartheta}{1-\vartheta} < \nu P$, it is possible that entry rises when $\Lambda > 0$. In that case we still have $\hat{P} < 0$ and $\hat{z} < 0$. But there exists a region where $\hat{e} > 0$.

A.4 Simple cases

This section discusses comparative statics in three simple cases: (i) linear entry costs $\psi(e) = \psi e$; (ii) quadratic entry costs $\psi(e) = e^2/2$; and (iii) concave entry costs $\psi(e) = \ln(e)$.

Linear entry costs. With linear entry cost. The free-entry condition (2.17) becomes $\nu P = \psi\vartheta/(1-\vartheta)$ implying a constant stock price. It follows directly that incumbent innovation is constant at $z = [\psi/(\nu\zeta\eta)]^{1/(\eta-1)}$. Entry e and labour used in production L^P are then jointly pinned down by the labour market clearing condition (2.19) and the value function (2.14). Any change in the stock price will shift the composition of production and entry but will not affect incumbent innovation.

Convex entry costs. Consider the case when $\psi(e) = e^2/2$. (A.21) simplifies to the expressions in (2.27).

Concave entry costs. Consider the case when $\psi(e) = \ln(e)$. The price equation in (A.21) can be written as:

$$\frac{\hat{P}}{P} = -\frac{(e + N\sigma^2)\left(\nu P - \frac{\vartheta}{(1-\vartheta)e}\right)}{\Lambda}, \tag{A.22}$$

where $\Lambda = \left(\nu P - \frac{\vartheta}{(1-\vartheta)e}\right)\left(\mu - z + \frac{(1-\vartheta)z}{\eta-1}\right) - \nu e\left[(1+\lambda)P + \frac{\vartheta}{e}\right]$ accounting for the fact that $\psi'(e) = 1/e$. As before incumbent innovation reacts proportional to the price $\hat{z}/z = \frac{1}{\eta-1}\hat{P}/P$ and the entry response is given by:

$$\frac{\hat{e}}{e} = \frac{(e + N\sigma^2)P\nu}{\Lambda}. \tag{A.23}$$

B Appendix – Extended Model with Heterogeneous Firms

This Appendix extends the model presented in Section 2 by introducing heterogeneity among incumbent firms. To keep the analysis simple I assume that firms face uncorrelated idiosyncratic risk but set aggregate risk to zero. Section B.1 describes the main changes in the extended model setup.

As the aggregate dynamics in the model with heterogeneous firms depend on the distribution of qualities, I derive the Kolmogorov-forward equation in Section B.2. This is non-trivial because the model features a discrete and finite number of firms, which implies that idiosyncratic shocks behave like aggregate shocks to the distribution, in particular shocks to individual firms change the weights across the firm size distribution for all firms. Finally, Section B.3 presents the model equilibrium.

B.1 Modified environment

B.1.1 Households and production

The structure of the household and production side remains unchanged as described in Sections 2.1.1 and 2.1.2.

B.1.2 Intermediate firms

Innovation. I introduce heterogeneity among firms via their R&D cost function. Following the standard setup in Akcigit and Kerr (2018) and Hobler and Matt (2024), firms innovate by investing $\zeta z_{it}^{\eta}\varphi^{\chi}$ of labour to generate a growth rate of their product quality at intensity z_{it}. The parameter χ captures the degree to which relative scale matters for the cost of innovation. An incumbent firm's innovation efficiency is:

$$\frac{\zeta z_{it}^{\eta} w_t}{z_{it} q_{it}} \left(\frac{q_{it}}{Q_t}\right)^{\chi} = \frac{\zeta z_{it}^{\eta-1} \omega_t}{\varphi_{it}^{1-\chi}}, \tag{B.1}$$

where $\omega_t \equiv w_t / Q_t$ is the wage-to-quality ratio. Equation (B.1) captures the quality improvement generated per unit of spending. When $\chi > 1$, larger firms closer to the frontier find it harder to innovate than laggards. When $\chi < 1$, the opposite is the case. The baseline model in Section 2 corresponds to $\chi = 1$. Firms' quality processes remain unchanged, see (2.5).

Stock returns. Stock returns are described by the process (2.6) with its coefficients as defined in (2.7).

B.1.3 Investors

Investors' problems remain the same as in the baseline model. Their demand functions are given by (2.10) and (2.13). As I discuss in the next section, the non-linear cost function introduces explicit price dependence of returns and therefore portfolio shares.

B.1.4 Stock price and investment

Firms maximise their stock price. In equilibrium, investors risk aversion gives rise to a risk premium. In the model with heterogeneous firms, this risk premium will depend on the firm size distribution. Imposing market clearing in each market, $\beta X_{it}^C + (1-\beta)X_{it}^M = 1$, yields the price function:

$$\max_{z_{it}} \left\{ \begin{array}{c} \pi(q_{it}) - \zeta z_{it}^{\eta} \varphi_{it}^{\chi} w_t + P'(q_{it}) z_{it} q_{it} + \frac{1}{2} P''(q_{it}) \left(\sigma_{it} q_{it}\right)^2 \\ - \left[r + \lambda \varsigma_{it}^2 + (1+\lambda) e_{it}\right] P(q_{it}) \end{array} \right\} = 0. \tag{B.2}$$

The firm maximises (B.2) by choosing its research intensity z_{it} taking into account how its decision affects the risk premium. The firm's first-order condition is given by:

$$\zeta \eta z_{it}^{\eta-1} \varphi_{it}^{\chi} w_t = P'(q_{it}) q_{it}. \tag{B.3}$$

B.1.5 Closing the model

The remainder of the model parallels the baseline case. Free-entry and growth are described by (2.17) and (2.18). The market clearing condition for the final good is (A.9), the market clearing conditions for bonds, intermediate goods and stocks are given by (A.11) to (A.13). The labour market clearing condition is:

$$L = L_t^P + \sum_{i=1}^{N} \psi(e_{it}) \varphi_{it} + \sum_{i=1}^{N} \zeta z_{it}^{\eta} \varphi_{it}^{\chi}. \tag{B.4}$$

B.2 Deriving the Kolmogorov-forward equation

In the model with homogeneous firms, all firms choose the same innovation intensity. Creative destruction rates are symmetric. Hence, aggregates do not depend explicitly on the distribution of quality levels across firms.

With heterogeneous firms, this is no longer the case. A Kolmogorov-forward equation (KFE) is needed to characterise the aggregate behaviour of the economy. It turns out that the model admits a stationary distribution in terms of firms' quality levels relative to the aggregate (or the mean). Given the vector $\boldsymbol{\varphi} = (\varphi_{1t}, \ldots, \varphi_{Nt})^T$, I denote the distribution of firms' relative quality levels by $p_t(\boldsymbol{\varphi})$.

Because there is only a discrete and finite number of N firms, deriving the KFE requires some additional steps compared to the baseline model or a model with a continuum of firms. In particular, shocks to individual firms now affect the distribution $p_t(\boldsymbol{\varphi})$ as they change the relative weights of firms $\boldsymbol{\varphi}$. Proposition 5 contains a description of the KFE and the step-by-step derivation.

Proposition 5 (Kolmogorov-forward equation with heterogeneous firms)**.** *Consider the model with heterogeneous firms described in Section B.1. Assume that the process for firms' product quality q_{it} follows (2.5) and aggregate quality Q_t is defined as in (2.4). Then, the Kolmogorov-forward equation for the distribution $p_t(\boldsymbol{\varphi})$ of firms' relative quality level $\boldsymbol{\varphi} = (\varphi_{1t}, \ldots, \varphi_{Nt})^T$ with $\varphi_{it} = q_{it}/Q_t$ is given by:*

$$\begin{aligned} \frac{\partial p_t(\boldsymbol{\varphi})}{\partial t} = &- \sum_{i=1}^{N} \frac{\partial}{\partial \varphi_i} \left[c_{it}(\boldsymbol{\varphi}) p_t(\boldsymbol{\varphi}) \right] + \frac{1}{2} \sum_{i=1}^{N} \sum_{\ell=1}^{N} \frac{\partial^2}{\partial \varphi_i \partial \varphi_\ell} \left[B_{i\ell t}(\boldsymbol{\varphi}) p_t(\boldsymbol{\varphi}) \right] \\ &+ \sum_{i=1}^{N} e_{it} \left[\left(\frac{\nu^N}{D_{it}^{N+1}} \right) p_t \left(\boldsymbol{J}_{it}^{-1}(\boldsymbol{\varphi}) \right) - p_t(\boldsymbol{\varphi}) \right], \end{aligned} \tag{B.5}$$

where $D_{it} = \nu - (\nu - 1)\varphi_{it}$. The remaining coefficients are given by:

1. *a drift term:*

$$c_{it}(\boldsymbol{\varphi}) = \varphi_{it} \left(\left[z_{it} - \overline{z}_t(\boldsymbol{\varphi}) \right] + \left[\overline{\Sigma}_t(\boldsymbol{\varphi}) - \varphi_{it} \sigma_{it}^2 \right] \right), \tag{B.6}$$

   with average innovation intensity $\overline{z}_t(\boldsymbol{\varphi}) \equiv \sum_{j=1}^{N} \varphi_{jt} z_{jt}$ and variance $\overline{\Sigma}_t(\boldsymbol{\varphi}) \equiv \sum_{j=1}^{N} \varphi_{jt}^2 \sigma_{jt}^2$.

2. *a covariance term:*

$$B_{i\ell t}(\boldsymbol{\varphi}) = \begin{cases} \varphi_{it}^2 \left[(1 - \varphi_{it}) \sigma_{it}^2 + \sum\limits_{j=1}^{N} (\varphi_{jt} \sigma_{jt})^2 \right] & \text{if } i = \ell, \\ \varphi_{it} \varphi_{\ell t} \left[\sum\limits_{j=1}^{N} (\varphi_{jt} \sigma_{jt})^2 - \varphi_{it} \sigma_{it}^2 - \varphi_{\ell t} \sigma_{\ell t}^2 \right] & \text{if } i \neq \ell. \end{cases} \tag{B.7}$$

3. *and a jump vector $\boldsymbol{J}_{it}^{-1}(\boldsymbol{\varphi}) = \left(J_{i1t}^{-1}, \ldots, J_{iNt}^{-1} \right)^T$ with elements given by:*

$$J_{ijt}^{-1} = \begin{cases} \frac{\varphi_{jt}}{D_{it}} & \text{if } j = i, \\ \frac{\nu \varphi_{jt}}{D_{it}} & \text{if } j \neq i. \end{cases} \tag{B.8}$$

Proof. The evolution of weights depends on a continuous component $d\boldsymbol{\varphi}^c$ and a series of discontinuous Poisson jumps $\boldsymbol{J}_{it}^{-1}(\boldsymbol{\varphi})$. First, starting with the continuous part, the definition $\varphi_{it} \equiv q_{it}/Q_t$ together with Itô's lemma implies that:

$$d\varphi_{it}^c = \frac{dq_{it}^c}{Q_t} - \frac{q_{it}dQ_t^c}{Q_t^2} - \frac{d\langle q_{it}, Q_t\rangle}{Q_t^2} + \frac{q_{it}d\langle Q_t\rangle}{Q_t^3}. \tag{B.9}$$

Using $dq_{it}^c = q_{it}\left(z_{it}dt + \sigma_{it}d\mathcal{Z}_{it}\right)$, $dQ_t^c = \sum_{j=1}^N q_{jt}\left(z_{jt}dt + \sigma_{jt}d\mathcal{Z}_{jt}\right)$, $d\langle q_{it}, Q_t\rangle = q_{it}^2\sigma_{it}^2 dt$, and $d\langle Q_t\rangle = \sum_{j=1}^N q_{jt}^2\sigma_{jt}^2 dt$ yields:

$$d\varphi_{it}^c = \varphi_{it}\Big(\left[z_{it} - \overline{z}_t(\boldsymbol{\varphi})\right] + \left[\overline{\Sigma}_t(\boldsymbol{\varphi}) - \varphi_{it}\sigma_{it}^2\right]\Big)dt + \varphi_{it}\left(\sigma_{it}d\mathcal{Z}_{it} - \sum_{j=1}^N \varphi_{jt}\sigma_{jt}d\mathcal{Z}_{jt}\right), \tag{B.10}$$

where $\overline{z}_t(\boldsymbol{\varphi}) \equiv \sum_{i=1}^N \varphi_{it}z_{it}$ and $\overline{\Sigma}_t(\boldsymbol{\varphi}) \equiv \sum_{j=1}^N \varphi_{jt}^2\sigma_{jt}^2$ as used in Proposition 5. Define the first term in parentheses as $c_{it}(\boldsymbol{\varphi}) \equiv \varphi_{it}\Big(\left[z_{it} - \overline{z}_t(\boldsymbol{\varphi})\right] + \left[\overline{\Sigma}_t(\boldsymbol{\varphi}) - \varphi_{it}\sigma_{it}^2\right]\Big)$ in (B.6) and note that the variance term can be written as $\sum_{j=1}^N G_{ij}(\boldsymbol{\varphi})d\mathcal{Z}_{jt} = \varphi_{it}\sum_{j=1}^N\left(\sigma_{it}\mathbf{1}_{i=j} - \varphi_{jt}\sigma_{jt}\right)d\mathcal{Z}_{jt}$, or equivalently:

$$d\varphi_{it}^c = c_{it}(\boldsymbol{\varphi})dt + \sum_{j=1}^N G_{ij}(\boldsymbol{\varphi})d\mathcal{Z}_{jt}. \tag{B.11}$$

As the liquidity shocks in the model are uncorrelated across firms, the diffusion covariance matrix between product lines is simply:

$$\boldsymbol{B}_t(\boldsymbol{\varphi}) = \boldsymbol{G}_t(\boldsymbol{\varphi})\boldsymbol{G}_t(\boldsymbol{\varphi})^T, \tag{B.12}$$

where the elements are defined in (B.7).

Second consider the jump part of the process for φ_{it} driven by creative destruction from outsiders that improve the quality of an existing product by a deterministic step size $\nu - 1$. Define $\boldsymbol{\Lambda}_{it}(\boldsymbol{\varphi})$ as the Jacobian of the shares related to the jumps. When one firm j experiences a jump, the aggregate quality level Q_t moves to $Q_t^+ = Q_t + (\nu-1)q_{jt} = \left[1 + (\nu-1)\tilde{\varphi}_{it}\right]Q_t$, where $\tilde{\varphi}_{it}$ is the quality ratio before the jump. Writing the post-jump shares as φ_{it}, the mappings between shares for firm i given a jump of j before, $J_{it}(\tilde{\boldsymbol{\varphi}})$, and after the jump, $J_{it}^{-1}(\boldsymbol{\varphi})$, are:

$$J_{it}(\tilde{\boldsymbol{\varphi}}) = \begin{cases} \frac{\nu\tilde{\varphi}_{it}}{1+(\nu-1)\tilde{\varphi}_{jt}} & \text{if } i = j, \\ \frac{\tilde{\varphi}_{it}}{1+(\nu-1)\tilde{\varphi}_{jt}} & \text{if } i \neq j, \end{cases} \quad \text{and} \quad J_{it}^{-1}(\boldsymbol{\varphi}) = \begin{cases} \frac{\varphi_{it}}{\nu-(\nu-1)\varphi_{jt}} & \text{if } i = j, \\ \frac{\nu\varphi_{it}}{\nu-(\nu-1)\tilde{\varphi}_{jt}} & \text{if } i \neq j. \end{cases} \tag{B.13}$$

To economise on notation, I write $D_{it} = \nu - (\nu - 1)\tilde{\varphi}_{it}$ and collect terms in the jump vector $\boldsymbol{J}_{it}^{-1}(\boldsymbol{\varphi}) = \left(J_{i1t}^{-1}, \ldots, J_{iNt}^{-1}\right)^T$. To find the Jacobian $\boldsymbol{\Lambda}_{it}(\boldsymbol{\varphi})$ needed for (B.5), collect terms in a $N \times N$ matrix:

$$\boldsymbol{\Lambda}_{it} = \frac{\nu}{D_{it}} \mathcal{I}_{N\times N} + \boldsymbol{u}_{it}\boldsymbol{e}_i^T, \tag{B.14}$$

where $\boldsymbol{e}_i = (0, \ldots, 0, 1, 0, \ldots, 0)^T$ has zeroes everywhere but the i^{th} position, and $\boldsymbol{u}$ is:

$$\boldsymbol{u}_{it} = \begin{cases} \frac{\nu(1-D_{it})}{D_{it}^2} & \text{if } j = i, \\ \frac{\nu(\nu-1)\varphi_{jt}}{D_{it}^2} & \text{if } j \neq i, \end{cases} \tag{B.15}$$

such that $\boldsymbol{u}_{it}\boldsymbol{e}_i^T$ is an $N \times N$ matrix. Intuitively, (B.14) has diagonal elements $\Lambda_{\ell\ell} = \nu/D_i^2$ for $\ell \neq i$, that is, the lines where no jump occurred but whose weight shifts due to a jump on another line i, and $\Lambda_{ii} = \nu/D_i$ for the row that corresponds to the element where the jump occurred. The off-diagonal elements are zero apart from the ℓ^{th} column with $\Lambda_{i\ell} = \nu(\nu-1)\varphi_{it}/D_i^2$ for $\ell \neq i$, reflecting how the jump changes the numerator of the share in (B.13).

To apply the change-of-variable formula when deriving the KFE, the determinant of this Jacobian is needed. We have:

$$\det(\boldsymbol{\Lambda}_{it}) = \det\left(\frac{\nu}{D_{it}}\mathcal{I}_{N\times N}\right)\left[1 + \boldsymbol{e}_i^T\left(\frac{\nu}{D_{it}}\mathcal{I}_{N\times N}\right)^{-1}\boldsymbol{u}_{it}\right] = \frac{\nu^N}{D_{it}^{N+1}}. \tag{B.16}$$

Finally, collect the drift (B.6), the covariance (B.7), and the Jacobian (B.16) to obtain the KFE in (B.5).

□

B.3 Equilibrium

We can now discuss the equilibrium in this economy. I will first formally define the growth path and then derive firms' equilibrium innovation.

B.3.1 Equilibrium definition

Definition 2. *The economy is on a stochastic balanced growth path if all aggregate variables – output Y_t, consumption C_t, profits Π_t, wages w_t, and intermediate production costs Y_t^P – grow at a common expected growth rate g_t, the growth rate of aggregate quality Q_t defined in* (2.18)*; the risk-free rate is constant at r, the risk-neutral household's discount rate defined in*

(2.1); *and stock returns follow the process in* (2.6). *Agents' optimisation problems satisfy the following conditions:*

(i) *households choose consumption and savings according to* (2.1)*;*

(ii) *the final good firm makes zero profits. Its demand functions for* $i = 1, \ldots, N$ *intermediate inputs and production labour are given by* (2.3)*;*

(iii) *intermediate firms set prices and quantities, and earn profits according to* (A.2)*; they maximise their stock price* (B.2) *by choosing the drift and volatility of their quality process* (2.5) *according to the first-order conditions* (B.3)*;*

(iv) *market-driven investors maximise* (2.9) *subject to the law of motion* (2.8) *by choosing their portfolio share of each risky stock* $i = 1, \ldots, N$ *according to* (2.10)*;*

(v) *risk-constrained investors maximise* (2.12) *subject to a law of motion equivalent to* (2.8) *and* (2.11) *by choosing the portfolio share of stocks* $i = 1, \ldots, N$ *according to* (2.13)*;*

(vi) *product quality follows the Kolmogorov-forward equation* (B.5)*;*

(vii) *and the free entry condition* (2.17) *holds;*

such that the markets for the final good, labour, the risk-free bond, N *intermediate goods, and* N *stocks described by* (A.9)*,* (B.4)*,* (A.11)*,* (A.12)*, and* (A.13) *clear.*

B.3.2 Solving for equilibrium

The firms' first-order condition (B.3) implies that innovation choices are no longer symmetric. Consequently, the model does not have a closed form solution. Nevertheless, there exists a stationary distribution of relative quality levels $p(\varphi)$ and the model can be solved numerically. A future version of this paper will contain the numerical solution under a suitable calibration.

C Appendix – Data

This Appendix supplements Section 3.1.1. Appendix C.1 presents descriptive statistics on the fund-level and firm-level data including standard balance tests. Appendix C.2 discusses the representativeness of the balance sheet data.

C.1 Descriptive statistics

This Appendix contains descriptive statistics for fund portfolios and further summary statistics on the distribution of the shift-share instrument constructed from those portfolios as well as standard balance tests.

C.1.1 Pension fund balance sheet data

Table C.1 reports summary statistics for the pension fund balance sheet data. In 2002/2003, the average pension fund in the sample held £1.814 billion in assets which grew to £3.455 billion in 2006/2007. The average funding level, defined as the ratio of a pension fund's market value of assets to the present value of future pension obligations, was 0.92 in 2001/2002 and 0.84 in 2006/2007. Hence, the average fund was *underfunded* as defined under the SFO. In fact, in 2002/2003 there were only 11 fully funded schemes.

In 2002/2003, the average portfolio share was 42 percent for U.K. equity and 25 percent for overseas equity. The domestic share well to 0.36 percent in 2006/2007 while the overseas share rose to 31 percent. For comparison, the average pension scheme (public and private) saw a decrease in its U.K. equity holdings from 38 percent to 25 percent over the same period. While the funds in the sample held larger shares of U.K. equities, the scale of the decline was similar.

C.1.2 Shift-share instrument

This Appendix contains summary statistics for the shift-share instrument used in the main specification in Section 3.3.1.

Shares. Identification in my setting comes from the shocks rather than the shares. As discussed in Borusyak et al. (2025) and Aghion et al. (2022, p.29), the effective number of shocks in this research design can be quantified by estimating the inverse Herfindahl-Hirschman index (HHI) of the shares $s_{i,f,2002}$. When a few pension funds hold large fractions of individual stocks, the effective sample size is small and the shift-share instrument will not be a consistent estimator. In my application, the inverse HHI of the shares 153, indicating that the effective sample size is large.

Table C.1: Fund data summary statistics pre and post reform

	FY 2002/2003					FY 2006/2007				
	Mean	Std	P25	P50	P75	Mean	Std	P25	P50	P75
Active members	39926.66	30815.05	15650.84	29177.24	50154.55	43165.88	32700.89	18597.26	32028.19	53180.54
Total members	79155.39	62613.12	30143.21	56863.26	99483.67	91786.12	70726.97	38234.66	65770.56	112547.11
Net assets	1814.20	1626.39	567.39	1097.59	2202.74	3455.45	3040.65	1139.79	2182.32	3887.51
Investments	1788.07	1600.46	557.79	1092.67	2188.54	3431.98	3011.88	1133.66	2168.70	3871.85
Funding level	0.92	0.12	0.85	0.92	0.99	0.84	0.08	0.79	0.84	0.89
Bonds	0.19	0.06	0.15	0.19	0.23	0.18	0.06	0.14	0.16	0.20
UK equity	0.42	0.06	0.39	0.43	0.46	0.36	0.06	0.33	0.36	0.39
Overseas equity	0.25	0.05	0.21	0.24	0.29	0.31	0.06	0.26	0.31	0.35
Property	0.07	0.04	0.05	0.07	0.10	0.08	0.04	0.06	0.09	0.10
Other	0.07	0.07	0.03	0.04	0.09	0.08	0.07	0.03	0.06	0.09

This table shows summary statistics for the 98 pension funds in my sample. The data are reported for the financial year 2001/2002, before the Pensions Act 2004 was announced, and the financial year 2006/2007, after the Act came into effect. Net assets and investments are reported in million pounds sterling. Bonds, equities, property, and others are reported as portfolio shares.

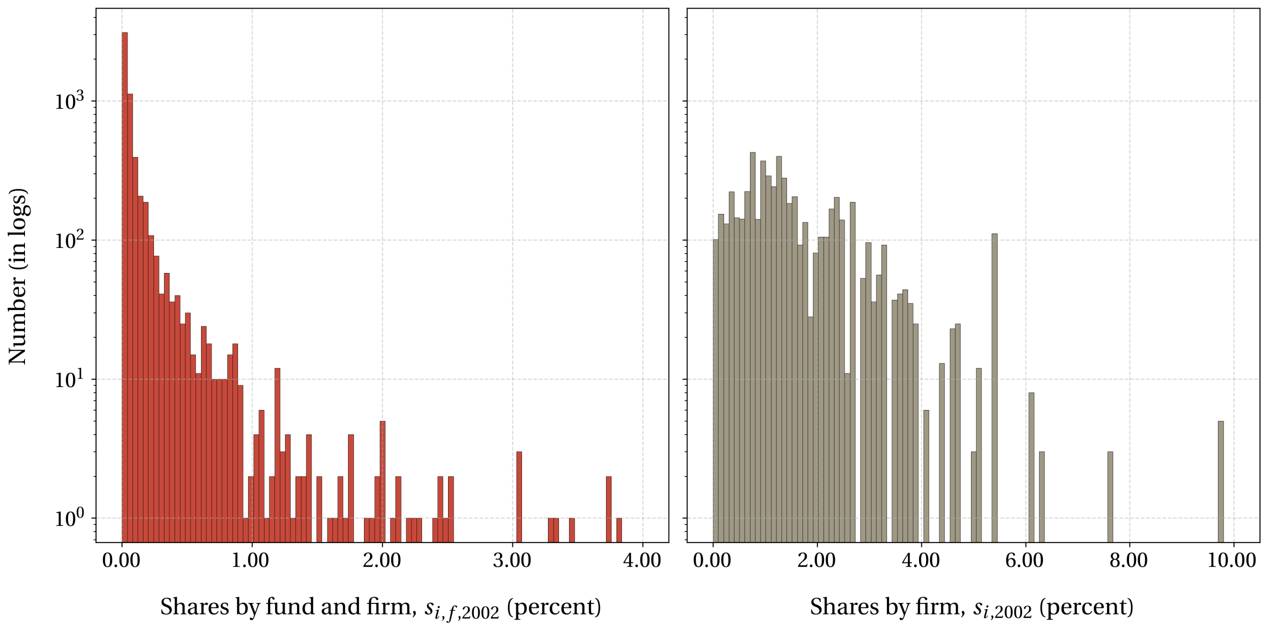

Figure C.1: Distribution of shares.

Figure C.1 plots the distribution of shares. The left panel contains a histogram of the shares across funds and firms, $s_{i,f,2002}$. The right panel shows the shares at the firm level, $s_{f,2002}$. A small number of outliers with shares large than five percent have been dropped from the plot for readability.

C.1.3 Firm-level balance tests

As discussed in the main text, the key threat to identification arises from unobservable correlation between the treatment, that is the divestment from equities at the fund level after the reform, and the outcome variables of interest, for example changes in R&D expenditure over assets or changes in capex over assets. An example for such a threat would be systematic assortative matching between firms and pension funds. This would lead to correlation between funds' divestment from equities and average firm performance. To further alleviate this concern Figure C.2 below provides standard balance tests. The figure illustrates that the key outcome variables R&D and capital expenditure pre reform are uncorrelated with the subsequent treatment, that is, the average firm-level divestment.

C.2 Representativeness

One might be concerned about the representativeness of the sample compared to other defined-benefit or defined-contribution pension schemes in the United Kingdom. The Office of National Statistics' (ONS) survey of pension funds, a repeated industry cross-section, can

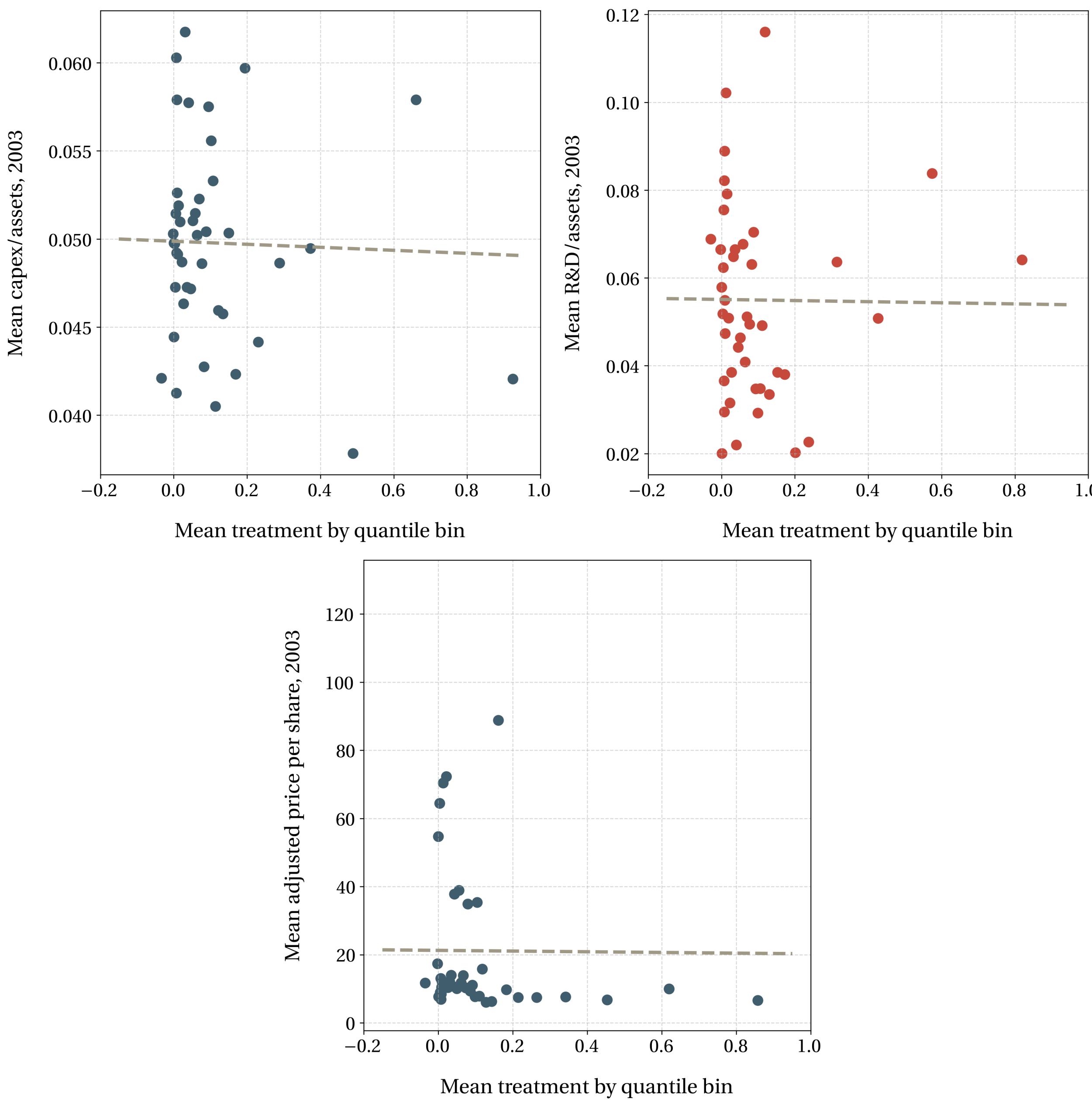

Figure C.2: Balance tests for capital expenditure and R&D spending

This graph shows a bin-scatter plot for pre-reform capital expenditure relative to assets (left panel) and pre-reform R&D spending, again relative to assets (right panel), as well as the line of best fit of the binned regression. The x-variable is the mean treatment by quantile bin, with the treatment $Z_{i,2002}$ defined as in regression 3.3, and the y-variable is the mean capital expenditure or R&D spending relative to assets, respectively.

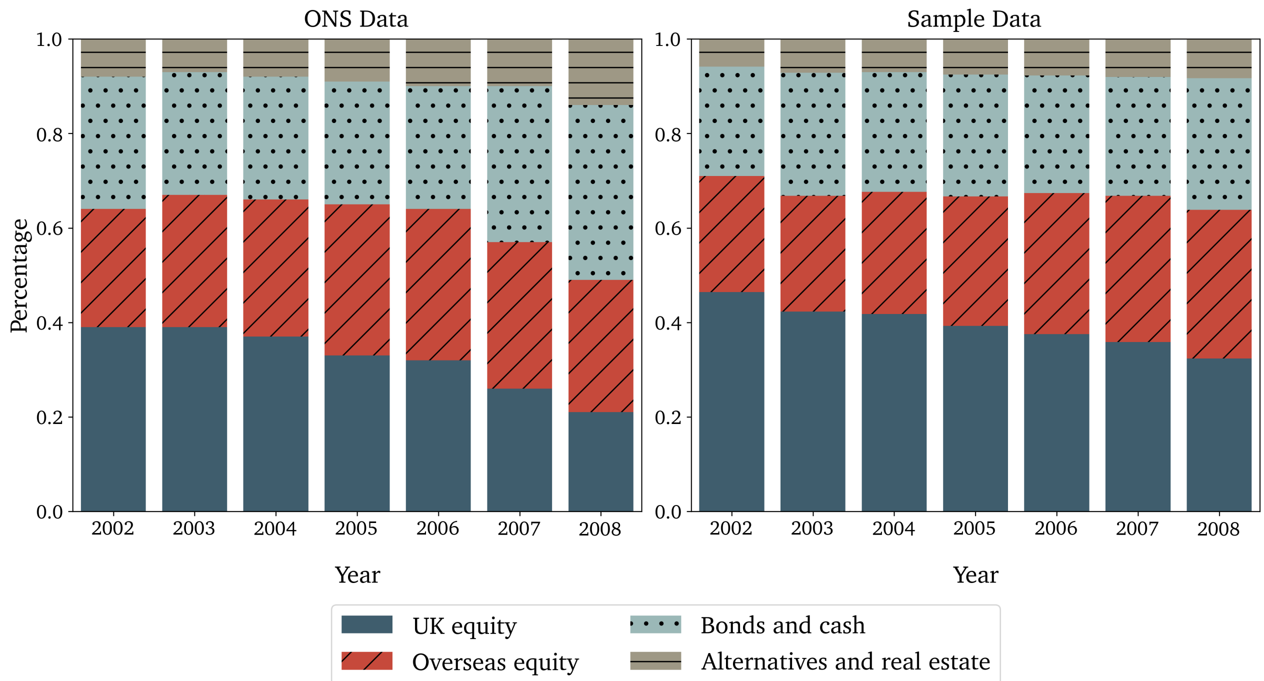

Figure C.3: Representativeness of asset allocation data

This graph shows the asset allocation of an average (size-weighted) pension scheme in the Office of National Statistics' (ONS) MQ-5 survey (left) and in the sample data (right).

be used to assess the representativeness of the sample portfolio allocation for the wider industry.

Figure C.3 shows the portfolio shares of the average fund in my sample (right) and the average scheme in the ONS data (left). On average the pension funds in sample hold a slightly larger fraction of their portfolio in equity and a smaller fraction in bonds than the average scheme. The trends following the pensions reform are similar: All schemes decrease their portfolio holdings of U.K. equities by around 12 percentage points, if anything the drop is slightly more pronounced for other schemes.

C.3 Portfolio concentration

Pension fund portfolios are relatively concentrated with little variation between funds. Table C.2 shows summary statistics on the make-up of pension fund portfolios. I report the average number of stocks held, the standard deviation, the HHI, the standard deviation of the HHI across funds, and the cosine similarity between portfolios for the years 2002 to 2006, which I define as the start and end date of the pension reform.

The average number of stocks held is 164 with standard deviation 65. The average number of stocks decreases to 139 over the period. The portfolio concentration is very low

Table C.2: Portfolio concentration

year	2002	2003	2004	2005	2006
$\mu(N)$	163.8	159.2	149.8	142.1	139.4
$\sigma(N)$	65.4	63.2	58.5	55.8	54.2
$\mu(HHI)$	0.1	0.1	0.1	0.1	0.1
$\sigma(HHI)$	0.018	0.018	0.018	0.020	0.020
μ(Cosine)	0.8	0.8	0.8	0.8	0.8
σ(Cosine)	0.3	0.3	0.3	0.3	0.3

This table shows summary statistics on pension schemes' portfolio concentration. $\mu(\cdot)$ and $\sigma(\cdot)$ denote the mean and standard deviation, respectively. The variables are the number of stocks N, the HHI, and the cosine similarity defined in equation C.1.

with an HHI of 0.06 and a standard deviation of 0.01. The cosine similarity, defined as:

$$\text{Cosine Similarity}_{f,f'} = \frac{\boldsymbol{\alpha}_f \cdot \boldsymbol{\alpha}_{f'}}{\|\boldsymbol{\alpha}_f\| \, \|\boldsymbol{\alpha}_{f'}\|} = \frac{\sum_{i=1}^{N} \alpha_{f,i} \, \alpha_{f',i}}{\sqrt{\sum_{i=1}^{N} \alpha_{f,i}^2} \sqrt{\sum_{i=1}^{N} \alpha_{f',i}^2}} \tag{C.1}$$

measures the pair-wise similarity between the portfolio vectors α_f and $\alpha_{f'}$ of any two funds f and f' for all funds $1, \ldots, F$ in the sample. A value of zero indicates no overlap between two portfolios a value of 1 indicates identical portfolios. The similarity between portfolios is high at 0.8 with standard deviation 0.3. There are no time trends in the concentration of pension fund portfolios or their similarity according to C.1.

D Appendix – Empirics

D.1 Trends in asset allocation

In this section, I document four *stylised facts* about the asset allocation of pension funds, which motivate my firm-level analysis: First, the pension reforms of the 1990s and early 2000s precipitate a long-run decline in the equity portfolio share on pension funds' balance sheets. Second, until the mid 2000s, this decline was driven by valuation effects on U.K. assets rather than active sales of stocks. Net quantities held by pension funds remained stable. Third, pension funds started to actively sell equities only with the announcement and implementation of the Pension Act 2004. And fourth, sell-offs were concentrated among those pension funds with large accounting deficits prior to the implementation of the reform.

D.1.1 Portfolio composition

Figure D.1 shows the portfolio composition for an average DB pension fund in the U.K. from 1962 to 2023. Until the early 1980s, pension schemes typically targeted an equity allocation of around 50 percent with the remainder held in gilts and property trusts.

Following the liberalisation of the 1980s, pension funds began to diversify internationally. Equity holdings reached a peak of almost 75 percent during the 1990s and remained relatively stable at around 60 percent until the early 2000s when falling equity prices after the end of the dotcom bubble brought down the equity portfolio shares. With the introduction of the Pensions Act 2004, as indicated by the dashed black line, the equity portfolio share started to decline and hit around 50 percent prior to the Global Financial Crisis (GFC). Within equity, the share of U.K. investment already started to decline in the late 1990s. The announcement and implementation of the Pensions Act 2004 accelerated this trend.

D.1.2 Prices versus quantities

Of course, the decline in the equity portfolio share could be driven by either active sales or by valuation effects of domestic stocks. To separate price from quantity effects, I run a decomposition in Appendix D.1.2. The analysis confirms that the reduction in pension funds' equity portfolio share prior to the announcement of the 2004 reform was driven by relative valuation effects but pension funds remained net buyers of U.K. equities. Post announcement, however, they became net sellers of equities with valuation effects even softening the decline in the equity portfolio share.

To disentangle changes in relative prices of U.K. equities compared to overseas equities from active sales of U.K. stocks, I introduce the following decomposition. Let E_t denote

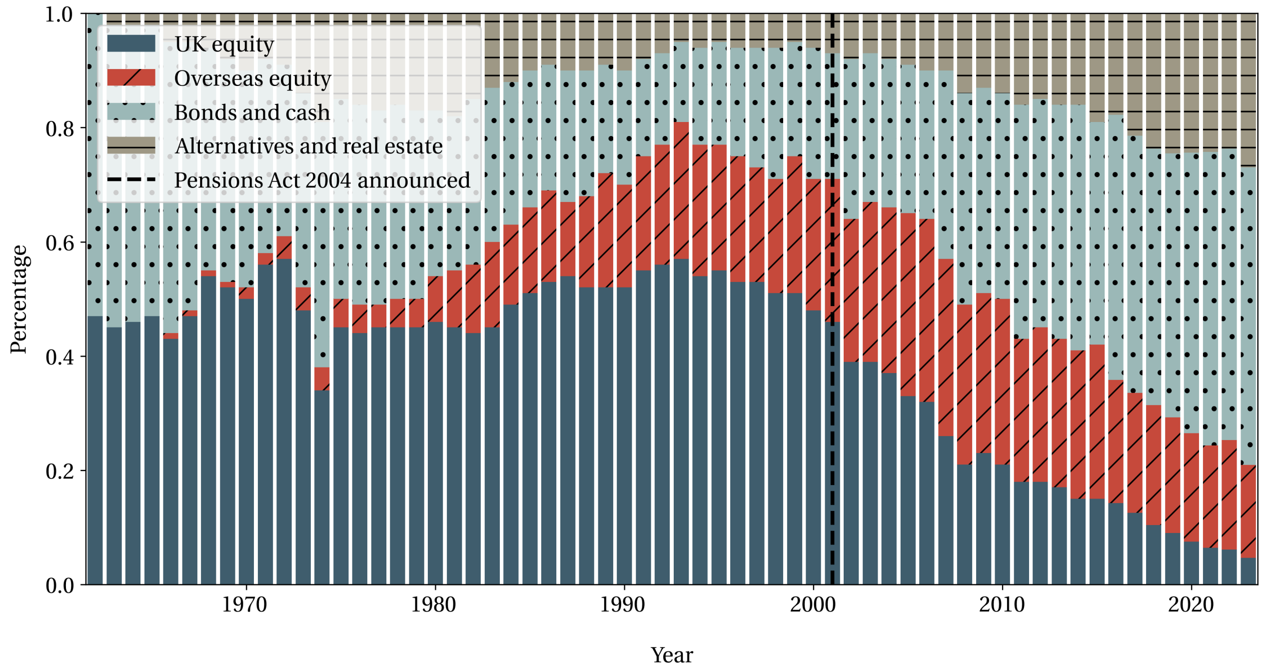

Figure D.1: Long-term trends in pension fund asset allocation.

This figure shows the long-term evolution of an average defined-benefit pension fund. The data in this figure is from the Office of National Statistics' (ONS) Financial Survey of Pension Schemes (FSPS), a repeated cross-section of 360 pension schemes in the U.K.. Averages are seize-weighted across pension funds to reflect average holdings across funds.

total value of holdings at time t consisting of quantities Q_t at prices P_t. The change in total holdings between years t and $t+1$ consists of a price component, a quantity component, and an interaction term:

$$E_{t+1} - E_t = \underbrace{Q_t\left(P_{t+1} - P_t\right)}_{\text{price component}} + \underbrace{P_t\left(Q_{t+1} - Q_t\right)}_{\text{quantity component}} + \underbrace{\left(P_{t+1} - P_t\right)\left(Q_{t+1} - Q_t\right)}_{\text{interaction term}}. \tag{D.1}$$

The quantity component reflects the adjustment to holdings due to sales or purchases of stocks, while the price component captures changes in holdings due to price changes. Following the same logic, I define the change in any variable X_t between any two years t and $t+T$ as $\Delta X_{t+T} \equiv X_{t+T} - X_t$, where $X \in \{E, P, Q\}$. Chaining up these changes over time and cancelling terms, $\Delta E_{t+T} = (E_{t+T} - E_{t+T-1}) - (\dots) - (E_{t+1} - E_t)$, we obtain the following expression for the total change in equity holdings over the horizon T:

$$\underbrace{\Delta E_{t+T}}_{\text{total change}} = \underbrace{\sum_{j=0}^{T} Q_{t+j}\Delta P_{t+j+1}}_{\text{price component}} + \underbrace{\sum_{j=0}^{T} P_{t+j}\Delta Q_{t+j+1}}_{\text{quantity component}} + \underbrace{\sum_{j=0}^{T} \Delta Q_{t+j+1}\Delta P_{t+j+1}}_{\text{interaction term}}. \tag{D.2}$$

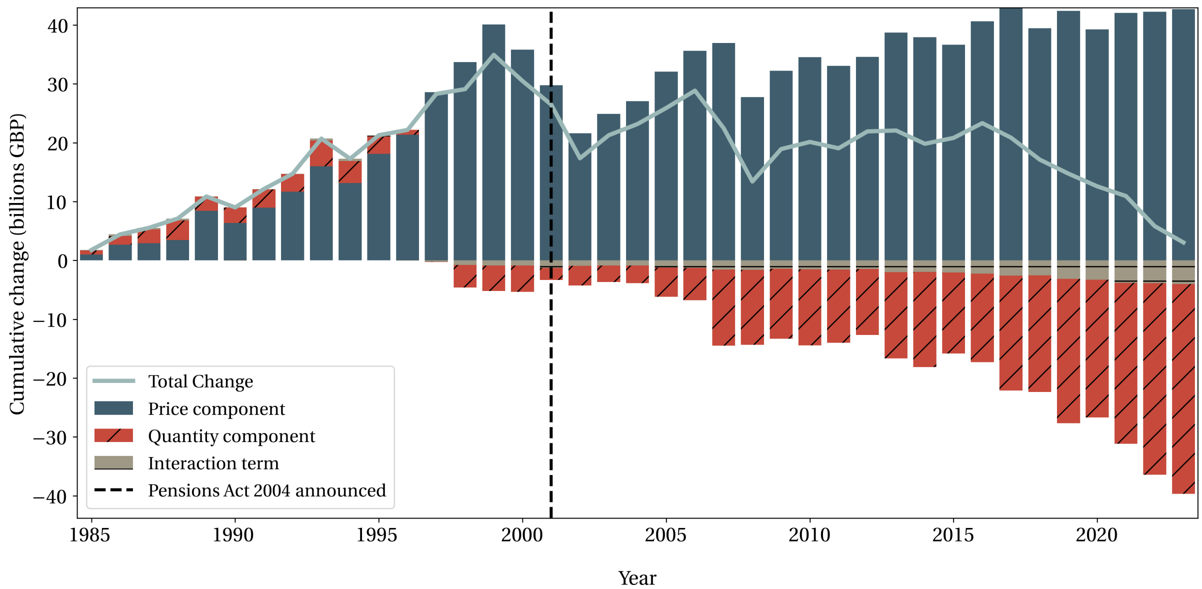

Figure D.2: Changes in stock holdings due to discretionary and price components.
This figure shows the cumulative changes in the equity portfolio share driven by prices, quantities, and their interaction corresponding to the decomposition in equation (D.2).

The decomposition in equation (D.2) consists of three parts. The first term captures the cumulative effect of stock price changes over time, holding quantities constant at their previous level. The second term reflects changes in the quantity of stocks held, holding quantities constant at their previous level. The third term is an interaction component that accounts for the combined effect of simultaneous changes in both prices and quantities. The price component can be thought of as reflecting the passive revaluation of existing holdings, while the quantity component reflects pension funds' active decisions to buy or sell stocks.

Figure D.2 illustrates this decomposition using the same holdings data used in the previous section. As in Figure D.1, the dashed black line indicates the announcement of the Pensions Act 2004. The solid teal line is the total change in equity holdings relative to the base year of 1985, blue bars correspond to price effects, red bars to quantity effects, and beige bars to the interaction component between the two.

Until the late 1990s, pension funds were net buyers of U.K. stocks with small-volume sales occurring during the financial years 1998 to 2000. Figure D.2 also illustrates that the Pension Act 1995 had relatively little if any effect on equity holdings. With the announcement and implementation of the Pensions Act 2004, pension schemes started to sell larger quantities of U.K. equity. This trend particularly accelerated after the financial crisis.

D.2 Aggregate flows for insurance companies and pension funds

This Appendix contains further plots illustrating the impact of the pensions reform. To show that equity sales limited to pension funds after the reform, Figure D.3 plots quarterly net flows for pension funds and insurance companies by asset class. Figure D.4 shows the cumulative flows.

As panel (1) illustrates, pension funds were net buyers of equities until early 2003 and only started to sell around the passage of the Pensions Act 2004. Insurance companies, in contrast, barely changed their equity holdings between 2002 and 2007. See Panel (2).

D.3 Fund-level results

This Appendix contains a series of results related to the cross-sectional variation at the fund level.

D.3.1 Funding regressions

This Appendix discusses the result of the main regression in (3.2). Column (1) of Table D.1 reports the coefficient of interest. The regression suggests that a one percentage point lower funding level in 2002 corresponds to a 0.2 percentage point additional decrease in the equity portfolio share for the period from 2003 to 2007.

To ensure that the effects are driven by quantities rather than prices, I rerun the analysis in Figure 3.2 with holdings normalised to 2003 prices. As these regressions are run on aggregate data, I use the chained price index from Appendix D.1.2 to deflate equity valuations to 2003 levels. The results of this adjusted quantity regression are in column (2). As with the raw equity valuations, the regression results suggest that pension funds with a lower funding level (a larger deficit) in 2002, sold larger quantities of equities in the period from 2003 to 2007.

D.3.2 Equity exposure regressions

This Appendix contains the regression output for the equity exposure regression (3.5) in Section 3.3.3. I report three variations of the same regression with no fixed effects, year, and year and fund fixed effects. The results suggest that a higher equity portfolio share is associated with a lower funding level. Controlling for the portfolio share, past returns do not predict pension schemes' funding levels.

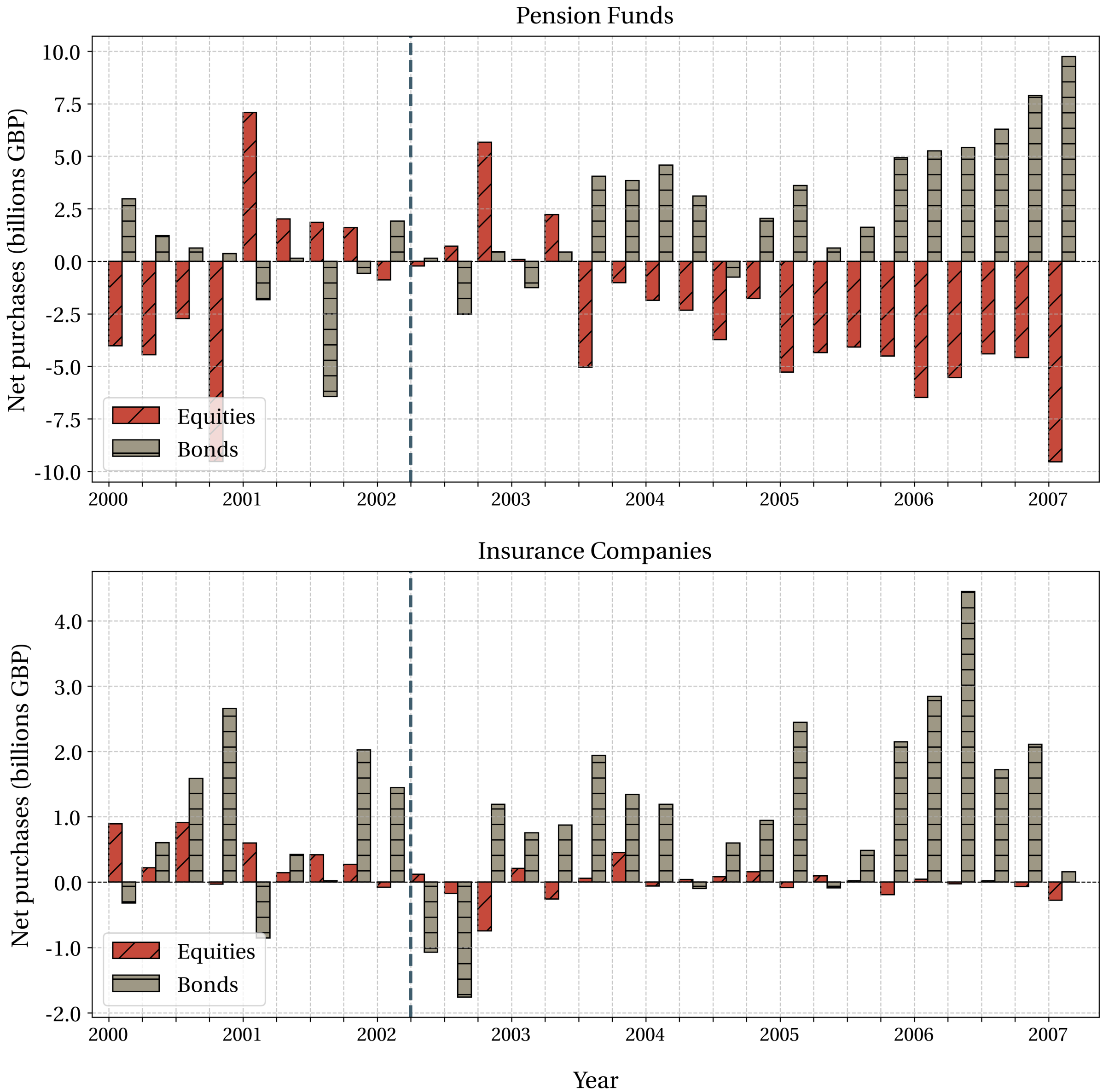

Figure D.3: Net purchases by pension funds and insurance companies.

The top (bottom) panel shows net purchases of bonds and equities for pension funds (insurance companies) from 2000-Q1 to 2007-Q1. The dashed line indicates the announcement of the Pensions Act 2004. The data come from the Office of National Statistics' MQ5 survey of pension funds.

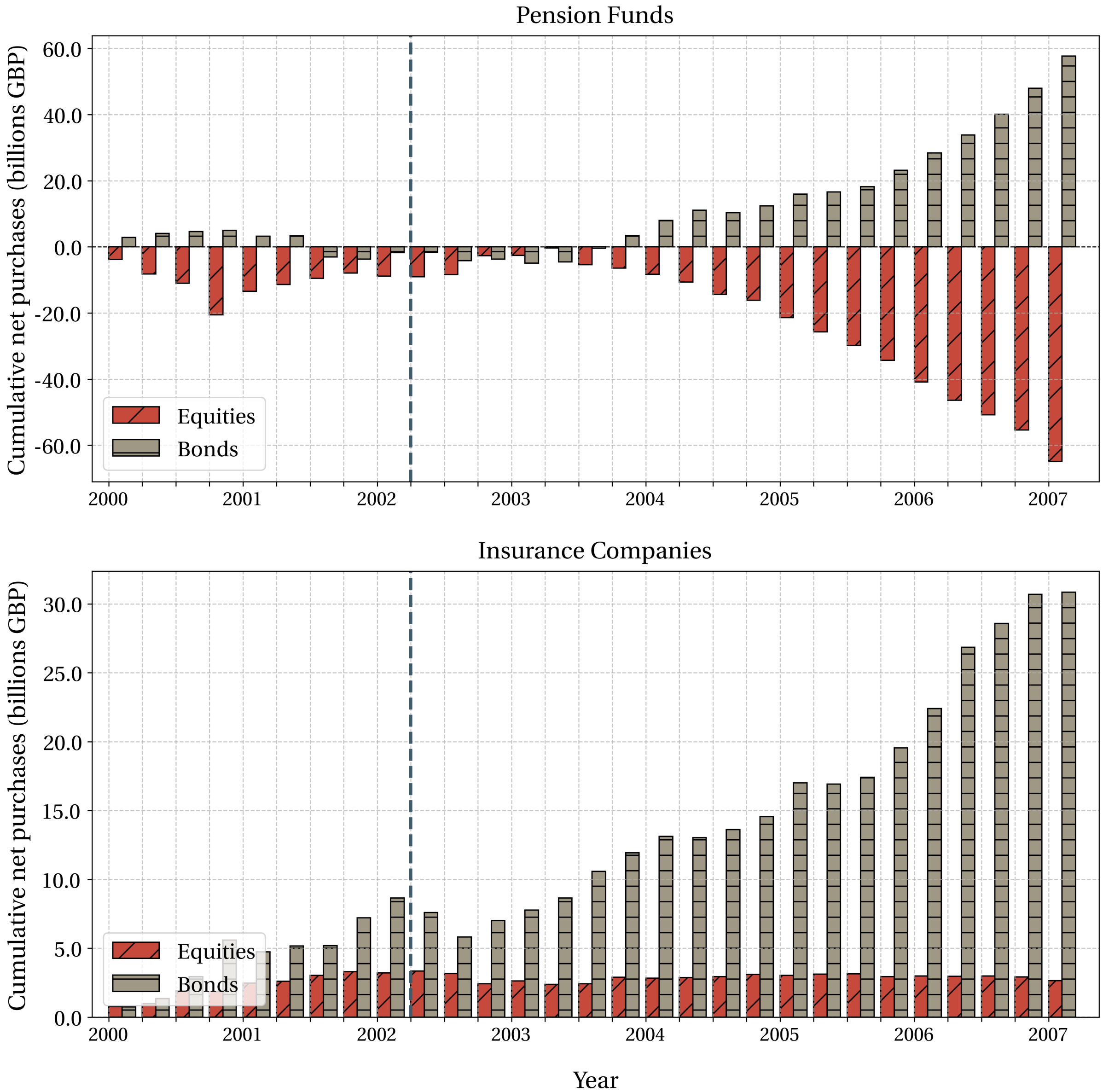

Figure D.4: Cumulative net purchases by pension funds and insurance companies.

The top (bottom) panel shows cumulative net purchases of bonds and equities for pension funds (insurance companies) from 2000-Q1 to 2007-Q1. The dashed line indicates the announcement of the Pensions Act 2004. The data come from the Office of National Statistics' MQ5 survey of pension funds.

Table D.1: Funding level and change in equity portfolio share

	(1)	(2)
	Δ (Equity Portfolio Share)$_{i,2007-2003}$	Δ (Adj. Equity Portfolio Share)$_{i,2007-2003}$
(Funding Level)$_{i,2002}$	0.1795***	0.4671*
	(0.0643)	(0.267)
Constant	-0.2281***	-0.4068**
	(0.0048)	(0.199)
Observations	98	98
R-squared	0.099	0.061

Robust standard errors in parentheses, $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

D.4 Firm-level results

This Appendix contains further empirical results on stock returns complementing the firm-level analysis in Section 3.3.

D.4.1 Leverage

Next, I evaluate the impact on firms' capital structure. I measure leverage as the ratio of long-term debt over the sum of equity and long-term debt. I then run a regression of the form (3.4) on the cumulative change in leverage based on firms' exposure to pension fund capital.

The top panel of Figure D.5 shows the cumulative effect of a one-percentage point reduction in pension fund investment on firms' leverage. An additional one-percentage point decrease in the share of pension fund investment increases leverage by about one percent after five years.

D.4.2 Asset maturity

Finally, I analyse how pension investment affects the composition and maturity of firm investment. Following Stohs and Mauer (1996) and in particular the methodology set out in Hubert de Fraisse (2024), I define a firm's asset maturity as:

$$AM_{i,t} = \frac{CA_{i,t}}{CA_{i,t} + NetPPE_{i,t}} \cdot 1 + \frac{NetPPE_{i,t}}{CA_{i,t} + NetPPE_{i,t}} \cdot \frac{1}{\delta_{i,t}}, \tag{D.3}$$

where $CA_{i,h}$ are firm i's current assets; $NetPPE_{i,h}$ is net property plant and equipment; and $\delta_{i,t}$ is the firm's depreciation rate. The idea (D.3) is to proxy for a firm's investment horizon

Table D.2: Equity exposure regression

	(1)	(2)	(3)
	Funding level	Funding level	Funding level
RoA	-0.00005***	-0.00001	-0.00001
	(0.00001)	(0.00001)	(0.00001)
Bond Share	-0.12696	-0.19010*	-0.09166
	(0.08348)	(0.07561)	(0.11179)
UK Equity Share	-0.16038*	-0.23401**	-0.19898*
	(0.07642)	(0.07826)	(0.09309)
Assets	0.00001***	0.00001***	0.00000
	(0.00000)	(0.00000)	(0.00001)
Membership	0.28929***	0.25701**	-0.21632
	(0.08480)	(0.07955)	(0.18056)
Constant	0.73052***	0.78644***	0.99960***
	(0.05927)	(0.06139)	(0.10255)
Year FE		✓	✓
Fund FE			✓
Observations	426	426	426

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

via the composition of its assets, or more precisely their depreciation rate. Current assets are used for production and therefore fully depreciated in a given financial year. Fixed assets, such as net PPE, can be used over a longer period, which is captured by their respective depreciation rates. Intuitively, the geometric depreciation weights $1/\delta_{it}$ assign high weights to assets with low depreciation rates which will be used to generate cash flow in the far future.[17]

Based on (D.3), I run the reduced-form regression (3.4) using asset maturity as the dependent variable. The results in Figure D.5 suggest that pension investment has a persistent effect on firms' asset composition. On average, firms more exposed to pension fund investment see a forwards shift in their asset composition. A one percentage point differential reduction in pension fund ownership is associated with a 0.4 percent decrease in asset maturity as defined in (D.3). This effect is highly persistent and still visible after eight to ten years. In line with my results for capital investment and R&D, I interpret these findings as corroboration of my hypothesis that firms react to the withdrawal of pension funds from equity markets by cutting down on long-term investment and focusing on the short term.

D.5 Robustness

This section presents the results of various alternative specifications and robustness checks complementing the main empirical results in Section 3.3. To confirm the robustness of the main specification, I rerun my main regression for series of alternative set-ups. In Section D.5.1 I swap out the base year 2002 against 2003. In Section D.5.2 I use an asset-under-management weighted shifter, where I weight sales by pension schemes' assets under management. In Section D.5.3 I construct a leave-one-out instrument that excludes fund f's sale of firm i's equity in the firm-specific shifter. Finally, in Section D.5.4 I leverage an alternative instrument based on pension schemes' membership structure prior to the reform.

D.5.1 Alternative base year

One might be concerned that the results in the main text are driven by shocks that occurred between the base period of 2002 and the implementation of the reform in 2004 and which were correlated with firms' exposure to pension fund equity prior to the reform.

To confirm the robustness of my results to this threat to identification, I rerun the main regressions in Section 3.3.1 using the alternative base year 2003 to construct the equity shares. Formally, let $s_{i,f,t}$ denote pension funds f's share of equity capital in firm i at time t for funds $f = 1, \ldots, F$ and let $M_{f,t}$ denote the total equity holdings of fund f on its balance

[17]The results are robust to using cash flow-based measures of asset maturity (Goncalves, 2021).

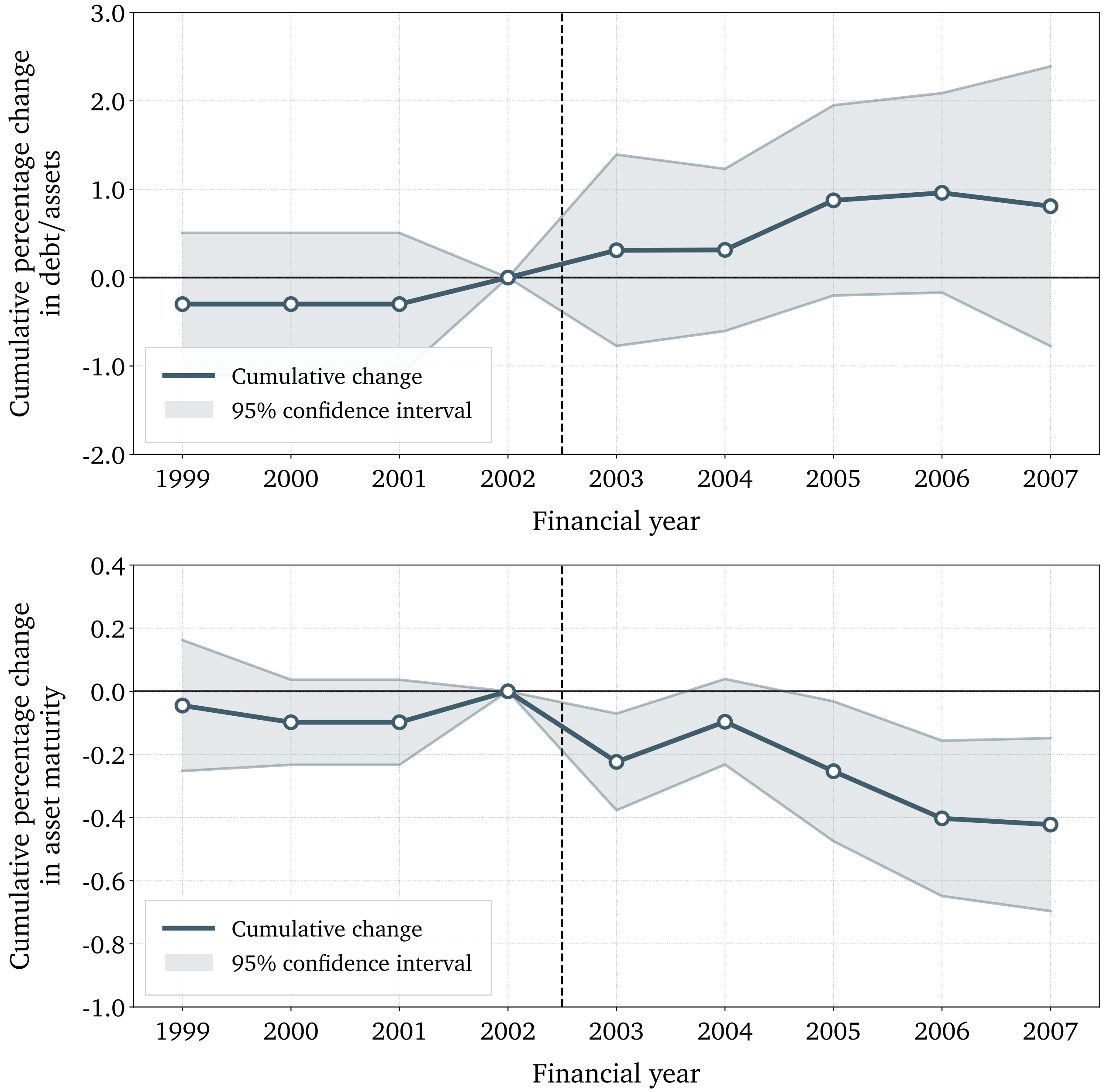

Figure D.5: Effect on leverage and asset maturity.

The top panel of the figure shows the cumulative effect of a one percentage point differential reduction in pension fund equity ownership at horizon h captured by β_h in equation (3.4). The dependent variable is leverage as defined in the text. The bottom panel of the figure shows the cumulative effect of a one percentage point differential reduction in pension fund equity ownership at horizon h captured by β_h in equation (3.4). The dependent variable is asset maturity defined in equation (D.3).

sheet at time t. I then construct the shift-share instrument as follows:

$$Z_{i,2003} = \sum_{f=1}^{F} \underbrace{s_{i,f,2003}}_{\text{2003-share}} \cdot \underbrace{(\ln M_{f,2003} - \ln M_{f,2007})}_{\text{2003-shift}}, \tag{D.4}$$

where I use the shift until 2007 as the baseline shock to retain the three-year structure of the instrument. I then run the reduced form (3.4) using the same controls as in the baseline and clustering standard errors at the firm level.

The results of my regression using this alternative specification support the baseline results with pre-treatment year 2002. The tables and figures are collated in the supplementary appendix available upon request.

D.5.2 Asset-weighted instrument

One might be concerned that the effects of the divestment from equity are related to the behaviour of a few large pension funds or that there is heterogeneity in divestment from equity based on fund size.

To confirm the robustness of my results to this threat to identification, I rerun the main regressions in Section 3.3.1 with fund weights based on pension schemes' assets under management (AUM) prior to the reform. Formally, let $s_{i,f,t}$ denote pension funds f's share of equity capital in firm i at time t for funds $f = 1, \ldots, F$ and let $M_{f,t}$ denote the total equity holdings of fund f on its balance sheet at time t. Moreover, I define the AUM-weights $a_{f,t}$ per fund as:

$$a_{f,t} = \frac{A_{f,t}}{\sum_{f=1}^{F} A_{f,t}}, \tag{D.5}$$

where $A_{f,t}$ are a pension schemes' total investment assets. The baseline period is again 2002, prior to the announcement of the Pensions Bill. Using these weights, I then construct the shift-share instrument as follows:

$$Z_{i,2002}^{AUM} = \sum_{f=1}^{F} \underbrace{s_{i,f,2002}}_{\text{share}} \cdot \underbrace{a_{f,2002} \cdot (\ln M_{f,2003} - \ln M_{f,2006})}_{\text{AUM-weighted-shift}}, \tag{D.6}$$

and rerun the reduced form (3.4) using the same controls as in the baseline and clustering standard errors at the firm level.

The results of my regression using the AUM-weighted instrument support the baseline specification. There is no evidence for heterogeneity based on fund size. The tables and figures are collated in the supplementary appendix available upon request.

D.5.3 Leave-one-out instrument

One might be concerned that the results in the baseline specification are driven by pension funds divesting large amounts from a few large firms. If the sell-off from these firms is large relative to other firms, the regression in (3.4) will be contaminated by a firm's "own shock" and the baseline specification would amount to a regression of a firms' stock price on an object that is highly correlated with its own price.

To confirm the robustness of my results to this threat to identification, I rerun the main regressions in Section 3.3.1 using a leave-one-out instrument (LOO). Formally, let $s_{i,f,t}$ denote pension funds f's share of equity capital in firm i at time t for funds $f = 1, \ldots, F$ and let $M_{i,f,t}$ denote the total equity holdings of fund f in firm i on its balance sheet at time t. I then construct the shift-share instrument as follows:

$$Z^{LOO}_{i,2002} = \sum_{f=1}^{F} \underbrace{s_{i,f,2002}}_{\text{share}} \cdot \underbrace{\left[\frac{\ln\left(\sum_{j \neq i} M_{j,f,2006}\right)}{\ln\left(\sum_{j \neq i} M_{j,f,2002}\right)} \right]}_{\text{LOO-shift}}, \tag{D.7}$$

where the shifter leaves out the shock to firm i itself. I then rerun the reduced form (3.4) using the same controls as in the baseline and clustering standard errors at the firm level.

The results of my regression using the LOO instrument support the baseline specification. There is no evidence that results are driven by firms' own shock. The graphs and tables are collated in the supplementary appendix available upon request.

D.5.4 Membership instrument

One might be concerned that the funding level itself is endogenous to firms' investment practices. I have already addressed these concerns in Section D.3.2 where I present evidence that suggests that differences in funding levels are driven by equity exposure, but not by differences in allocations within equity. Regardless, this Appendix provides an alternative set-up to the baseline regression (3.4) using an alternative instrument based on firms' membership structure.

The idea behind this instrument is as follows: The funding level $\mathcal{F}_t$ defined in (3.1) depends on the net present value of future pension obligations. How high a scheme's future obligations are depends, in turn, on the ratio of contributors to pensioners. A pension scheme with few active contributors and many pensioners faces higher liabilities while collecting lower contributions. Hence, its funding level should be systematically lower. If the differences in funding levels between pension schemes are indeed driven by their future pension obligations, the membership should be predictive of schemes' divestment from equities in response to the pensions reform. Figure D.5.4 illustrates that this is indeed the case. Using this logic, I define a scheme's age-structure:

$$N_{f,t} = \frac{(\text{Number of Active Contributors})_{f,t}}{(\text{Total Number of Members})_{f,t}}. \tag{D.8}$$

One interpretation of D.8 is that of a dependency ratio, that is, the number of active contributors per retiree or deferred member of the scheme. I then use the membership structure $N_{f,t}$ to construct the following instrument:

$$Z_{i,2002}^{MEM} = \sum_{f=1}^{F} \underbrace{s_{i,f,2002}}_{\text{share}} \cdot \underbrace{(\ln N_{f,2004} - \ln N_{f,2002})}_{\text{MEM-shift}}, \tag{D.9}$$

where I use the pre-reform trends in fund membership structure as an instrument for the funding level. I choose as the baseline period the years 2002 to 2004. The key assumption here is that the changes in membership structure are plausible exogenous to firm performance and affect the divestment from equities only through their impact on funding levels. I include the same control variables as before but do not cluster standard errors. I also control for fund size as Figure D.5.4 suggest that smaller funds tend to have higher dependency ratios.

I then rerun (3.4) using this instrument. The results from this alternative specification using the membership instrument are very similar to the baseline but with considerably less power due to the nature of the instrument. The graphs and tables are collated in the supplementary appendix available upon request.

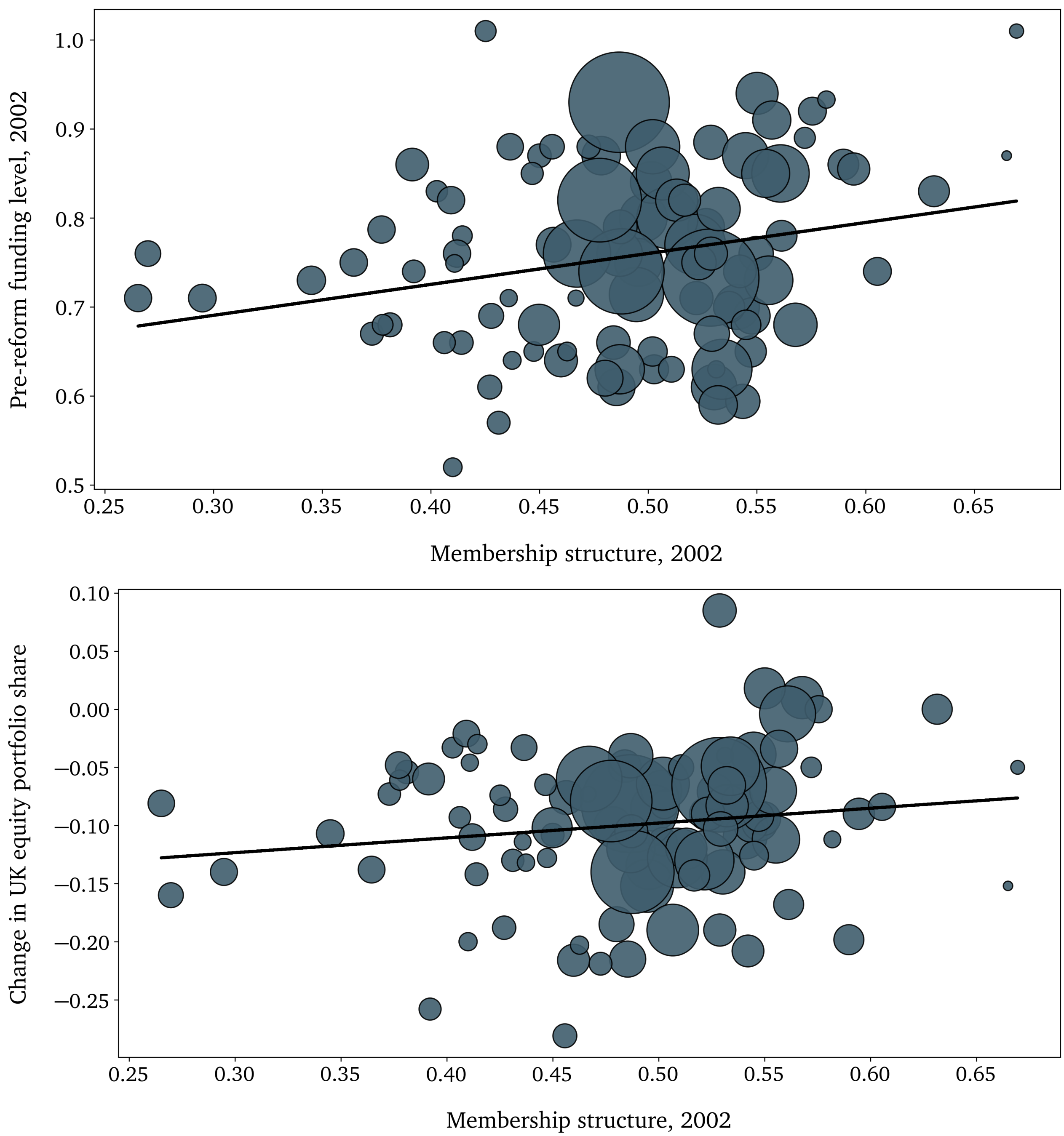

Figure D.6: Pre-reform membership structure, funding level, and post-reform equity sales.

The top panel shows the correlation between schemes' membership structure as defined in (D.8) and their pre-reform funding level, both measured in 2002. The bottom panel shows the correlation between membership structure and the change in the equity portfolio share after the reform from 2003 to 2007. These results are robust to removing outliers with a membership ratio $N_{f,2002} \leq 0.4$ and to seize-weighting schemes by assets.

E Appendix – Quantitative Model

E.1 Calibration procedure

This Appendix contains further notes on the calibration procedure in Section 4.1. The model features twelve structural parameters as listed in Table 4.1. I set $L = 1$ and $\kappa = 1$ to normalise the size of the economy and the initial penalty on pension funds. Moreover, we have $\eta = 2.5$, $\vartheta = 0.6$, and $r = 0.05$. I then set $\beta = 0.25$ such that, given a portfolio share $\alpha = 0.47$, the implied shareholding of risk-constrained investors is $\beta X^C/[(1-\beta)X^C+\beta X^M] = \beta\alpha^C W_t^C/Q_t \approx 0.12$, see (2.20). Next, the equilibrium portfolio share is $\alpha^C = \lambda/(2\kappa + \gamma)$. Using the definition of $\lambda = \gamma(2\kappa + \gamma)/[2\kappa(1 - \beta) + \gamma]$, one can invert the portfolio share expression to back out the value of γ that supports the targeted portfolio share given the other parameter values:

$$\gamma = 2\kappa(1 - \beta)\left(\frac{\alpha^C}{1 - \alpha^C}\right). \tag{E.1}$$

Having set the externally-calibrated parameters, the vector of moment conditions for the internal calibration is:

$$\mathcal{M}(\Gamma) = [g, \quad e, \quad \varepsilon, \quad m]^T, \tag{E.2}$$

where $m = \mu - r - e$ is the risk premium and $\varepsilon \equiv -N^{-1}\sum_{i=1}^{N} d\ln\left(\zeta z_{it}^{\eta}\omega\right)/d\kappa$ is the elasticity of innovation expenditure with respect to the change in investor composition as defined in (4.3), respectively. Given the theoretical moments in (E.2), I run the pattern search algorithm as described in (4.2). The parameters are $\{\sigma_a, \nu, \psi, \zeta\}$. I use a weighting matrix $\boldsymbol{W} = [1, 1, 1, 1]$ and the Euclidean norm $p = 2$.

E.2 Untargeted moments

To provide a sense of the model's quantitative fit, I report a series of untargeted moments implied by the baseline calibration. The model-implied average annual return on firms' stock μ is 15.76 precent. The risk premium is approximately 6.7 percent.

The implied annual volatility of stock prices is 20.6 percent, which is approximately 5.96 percent on a monthly basis. Damodaran (2019) reports a monthly standard deviation of risk premiums of 5.78 percent for the period between 1981 to 2001, which is 20.02 per year. In the model, the risk premium is split between an idiosyncratic and a creative destruction component. In the baseline calibration, 95 percent of stock price volatility comes from creative destruction risk and the remaining 5 percent from firms' idiosyncratic cash flow volatility driven by their liquidity shocks.

At the firm level, the model implies a price earnings ratio of $P_t q_{it}/(\pi_t q_{it} - \zeta z^\eta w_t \varphi_{it})$ of approximately 10.4, which is at the lower end of estimates for the U.K. economy in the late 1990s and early 2000s.

Labour used in the production of the final good is $L^P = 0.85$, which implies that around 15 percent of workers are employed in the research sector. Total research spending is defined as $\sum_{i=1}^N (\psi e_{it} + \zeta z_{it}^\eta) \varphi_{it} w_t / Y_t$ and accounts for 22 percent of GDP, which is much larger than in the data but not surprising given that the model is a Schumpeterian growth model that abstracts from capital accumulation.

E.3 Output and risk premium

Figure E.1 shows the behaviour of output and the risk premium, which has been omitted from the main text. The left panel shows output Y as a function of the regulatory constraint parameter κ. Tighter regulation on risk-constrained investors raises contemporaneous output through its impact on incumbent firms' labour demand and hence the equilibrium wage. When incumbent firms cut back on investment, the equilibrium wage drops, allowing intermediate producers to increase their output. In the baseline calibration, however, output is relatively inelastic with respect to changes in the regulatory constraint. This is because most labour is used for production already, $L^P = 0.85$.

The right panel of Figure E.1 shows the risk premium, defined as $\mu - r - e$, as a function of the regulatory constraint parameter κ. The risk premium is increasing in κ in the neighbourhood of the parameter changes considered in the model, but can locally decrease when firms overreact to changes in market risk aversion as explained in the Section 2. In the baseline calibration, is true for small values of $\kappa < 0.6$. This result is driven by changes in firm entry (the reallocation effect dominates).

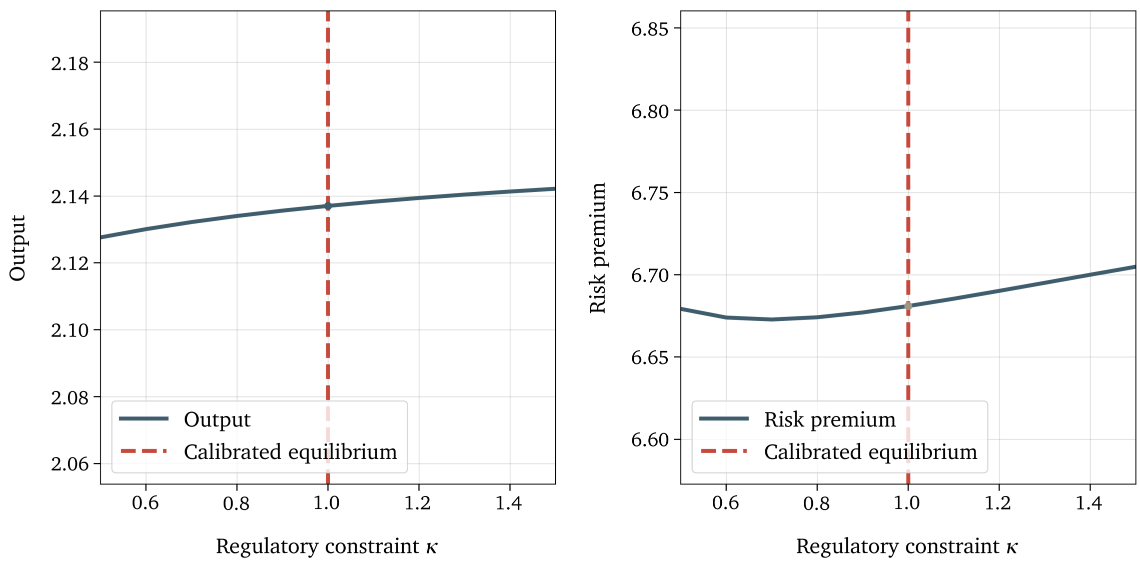

Figure E.1: Equilibrium output and risk premium.

The figure shows output (left) and risk premia (right) as a function of the regulatory constraint parameter κ. The red line denotes the calibrated pre-reform steady state.